

# Regimes' Characteristics and Time Series Forecasting

FGV-EESP Masters' Thesis

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This thesis investigates how regime-switching (RS) models learn and forecast under different data-generating processes, and whether each regime's distribution helps explain and predict model performance. I introduce a framework that separates the *series-generating process* (SGP) from the *regime-generating process* (RGP), and I formalize *regime-conditional metrics* (RC) that summarize differences between regime distributions. A Monte Carlo setup generates series, estimates models, and computes RC metrics. The framework is expandable, but I focus on: stationary  $AR(1)$  series; two-regime Markov Switching (MS), Self-Exciting Threshold (SET), and Smooth Transition (ST) mechanisms; MS, SET, and ST models, plus K-Means (KM) and Random Forest (RF); RC metrics based on the mean, SD, and lag-1 ACF. Results show that RGPs and SGPs interact in non-obvious ways, but RC metrics can sometimes characterize that behavior. KM and RF are the best performers, followed by SET and ST, while MS is more flexible and performs better in asymmetric regimes; KM and ST are robust across DGPs. Mis-specifying the RGP increases RMSE by 0.52. ST performs poorly on no-RS series, but other model-RGP interactions are generally insignificant. Matching the regime is important for performance, but not for KM. SET and ST perform best on series with high mean separation; ST performs worse with high ACF separation, and KM fares well when regimes differ in SD. Under-specifying the number of regimes is less harmful when regime distributions are minimally separated, whereas over-specifying is less harmful in the opposite case.

**Keywords:** Time series, Regime Switching.

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# 1 Introduction

Regime-switching (RS) models describe time series whose behavior – parameters – varies across regimes. They capture nonlinearities and are widely used in economics and finance, for instance, to model business cycles and market volatility. There are several regime-switching approaches, including stochastic ones, such as Markov-switching models, and deterministic ones, such as threshold models.

As with any forecasting model, it is important to understand the factors that drive performance, and how econometricians can use this knowledge to improve their models. In this work, I focus on: (i) these models’ ability to learn and generate accurate forecasts under mis-specification; and (ii) how that ability relates to the regimes’ distributions.

The first focus is common in forecasting econometrics: exactly identifying the data-generating process (DGP) is the exception, not the rule, so the modeling goal is to find a robust approximator. I therefore document how each RS model behaves under mis-specification, explore candidates for universal approximation, and study how each DGP component shapes the models’ learning problem.

The second focus is less orthodox and specific to RS models. These models aim to identify not only the series but also its regimes, allows describing each regime’s distribution and how they differ from each other. This characterization of regimes’ distributions might be informative for model performance. For example, if the DGP implies different intercepts across regimes, a model whose identified regimes share the same conditional mean likely misses that dynamic; conversely, a model may capture level shifts but fail to capture changes in persistence. These examples might seem obvious, but I show that this perspective yields useful information.

The nature of this project is exploratory. I simulate a diverse set of DGPs and look for stylized facts about how each RS model adjusts to them, and how the characteristics of the estimated regimes relate to that adjustment. In the remainder of this section I synthesize the methodology, describe the patterns I look for, and present some of the actual findings.

## 1.1 Basic methodology and hypothesis

The first step is to establish a theoretical framework that describes RS models in a unified way. Here, I denote the separate ‘ingredients’ in an RS DGP: the *series generating process* (SGP), the *regime generating process* (RGP), and what changes across regimes, the *regime nature* (RN). By varying these ‘ingredients’, one can define a diverse set of DGPs to study. I define the notion of regime-conditional (RC) metrics and discuss the different ways they can be computed and compared. I propose a Monte Carlo setup to generate series, estimate models, and calculate RC metrics. The code is available and implemented in a similarly modular and expandable way.

Developing a general framework was a goal in itself, but for this work, I consider only a specific set of DGPs, models, and metrics, answering only some of the questions the framework allows.

I focus on stationary  $AR(1)$  processes, with regime switching via Markov Switching (MS), Self-Exciting Threshold (SET), and Smooth Transition (ST), with symmetric and asymmetric variations. The regimes can differ in one of the three parameters of the  $AR(1)$ , and the change can be ‘big’ or ‘small’. I include a no-RS DGP as a baseline. Each RGP is accompanied by its related model, with the addition of an unsupervised RS K-Means (KM) model, and a Random Forest (RF). The RC metrics are based on the average, standard deviation (SD), and lag-1 autocorrelation (ACF), conditional on each regime.

For the analysis, I begin with an exploratory step to understand how the regimes’ distributions differ across these DGPs. Do the metrics correctly identify differences between distributions? Can they yield stylized facts that help clarify how these models work?

I then ask whether the distribution of regimes informs something about model performance. That is, do models fare better when faced with a specific profile of regimes' distributions? Does matching the distributions improve performance? Does under-specifying the number of regimes matter less if the regimes' distributions are similar, and vice versa?

For the more orthodox objectives, I first explore the general performance of the considered models in this specific pool of DGPs. Which one performs better? How do the models' components – fit, parameters, generated regimes – relate to that performance, and how does it vary across DGP scenarios (e.g. asymmetric DGPs)? Does mis-specifying the DGP impact performance, and how? Which model component is more important to match, in terms of performance?

The goal of this work is to answer these questions, generating stylized facts about the DGPs, and practical recommendations for modeling regime-switching time series. Results are conditional on the population of DGPs considered, and including more metrics could increase the explanatory power of the study.

Notable findings include: the interaction between RGPs and RNs is paramount to the behavior of the series; such behavior can be profiled by the dispersion of the RC metrics, but only with MS RGPs, SET and ST possibly requiring more metrics; MS models are more flexible and deal better with scenarios such as asymmetric RGPs, but SET and ST are otherwise better overall, while KM and RF are clearly on top; mis-specifying the RGP increases the RMSE by 0.52, but RGP interactions are insignificant.

SET and ST are best on series with high mean separation, ST does worse with high ACF separation, and KM fares well when regimes differ in volatility. However, this analysis is sensitive to the estimation error of metrics and varies across non-observable DGP factors, requiring further investigation. Under-specifying the number of regimes is attenuated in minimally separated regimes' distributions, and over-estimation is attenuated in the opposite case.

The rest of this work is divided as follows: Section 2 presents the literature review. The general framework is presented in Section 3 and Section 4, while the specific implementation chosen is presented in Section 5. The results are split into a more exploratory section (6) and a systematic one (7). Finally, Section 8 concludes.

## 2 Related literature

The regime-switching literature is vast and includes many model variations. In this section I map the models, their similarities, and their differences. I also summarize what is known about forecasting performance and the factors that drive it. Before that, I delineate the scope of the RS literature by discussing two closely related literatures.

The first is the state-space (SS) literature, exemplified by Kalman (1960). Although RS and SS models developed as largely independent fields, one can view RS as a subset of SS, where the state (regime) variable and the observed series variable are modeled separately. This separation is central to the framework used in this paper. Bridges between the literatures include switching state-space models and the seminal work by Kim (1994), which extends Hamilton’s Markov-switching model to general state-space models.

The second is the structural break (SB) literature. A natural starting point is Chow (1960). Much of this literature is devoted to diagnosing breaks, which also arise in RS settings through non-constant parameters. However, SB models typically treat breaks as exogenous and non-recurring. Bridging the gap, Bai; Perron (1998) allows for multiple unknown breaks, which can be relevant in RS contexts, while Chib (1998) shows that SBs can be formulated as Markov-switching processes that assign positive probability to staying in the initial regime and switching to the next, but not to switching back. Ultimately, I found that SB models were not comparable with RS models for the analysis done in this work, but future works could explore these connections.

### 2.1 Regime-switching models

Two aspects distinguish the main RS approaches: (i) whether the latent regime variable is modeled deterministically or stochastically, and (ii) whether changes between regimes are abrupt or smooth.

The most common deterministic models are threshold-based: an observable variable crossing one or more thresholds determines the regime. The work of Tong (1978), (1980) popularized the threshold autoregressive model, in which each regime has its own set of autoregressive parameters. Tong argued that capturing smooth transitions between regimes could matter, and Terasvirta; Anderson (1992), (1994) introduced the smooth transition autoregressive model, where the distance between an observable variable and a threshold determines each regime’s continuous weight.

On the stochastic front, the Markov-switching literature begins with Hamilton (1989) and the MSAR model, where an unobservable Markov process governs the regime – the probability of switching is time-invariant and depends only on the current regime. This implies a geometric distribution for the length of a regime instance.<sup>1</sup> The Markov-switching smooth transition model, as defined by Elliott; Siu; Lau (2018), exists, but it is more complex than the natural jump from the TAR to the STAR model.

#### 2.1.1 Variations of the classic models

Subsequent work introduced many variations in modeling the regime variable. The threshold variable can have a delay or a transformation; it can be the series itself, an exogenous variable, or a nonlinear combination of variables (Chen; Liu; So, 2011). Markov-switching models were extended to allow non-geometric distributions for time spent in a regime, and dependence on additional lags (Ferguson, 1980). The smooth transition function also admits several options, with common choices being logistic

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<sup>1</sup>Throughout this document, ‘regime instance’ will be used to describe a contiguous period of time without switches. In a given series, a given regime can have several instances.

and exponential functions. The models were generalized to any number of regimes. Medeiros (2000) shows that the STAR model is equivalent to a neural network in which the  $S$  regimes are  $S$  nodes in the hidden layer, a fact that will help describing its performance.

Some work blur the line between deterministic and stochastic models. Chang; Choi; Park (2017) uses threshold dynamics but introduces an innovation term that depends on the previous state’s innovation, and it simplifies to an MS model when the threshold variable is exogenous and stationary. Wu; Chen (2007) proposes a hybrid regime process, combining threshold and random switching. There are also unsupervised estimation approaches that make no assumptions about the latent process; for example, Akioyamen; Tang; Hussien (2020) uses clustering to identify regimes and then estimates the functional form separately within each regime, motivating the K-Means model considered in this work.

I defend that the functional form across regimes and the regime process itself are largely independent. This suggests further variants based on more complex within-regime dynamics than simple autoregression. ARMA models have regime-switching counterparts (Brockwell; Liu; Tweedie, 1992). One can model not only the mean but also the variance: the ARCH/GARCH family, widely used in finance, has regime-switching versions (Hamilton; Susmel, 1994), (Chen; Liu; So, 2011). More recently, models such as decision trees have been adapted to the regime-switching context, as in (Adam; Ötting; Michels, 2024). There are also extensions to multivariate time series.

General reviews on RS models include (Tan; Wu, 2025), (Potter, 2000), and (Hamilton, 2020). (Chen; Liu; So, 2011) focuses on threshold models, (Dijk; Teräsvirta; Franses, 2002) on smooth transition, and (Song; Woźniak, 2021) on Markov-switching. RS models appear in both frequentist and Bayesian frameworks, with Bayesian approaches drawing heavily on the state-space estimation literature. A mathematical definition of the models is present in Appendix A.

## 2.2 Forecasting performance

There are many research topics in RS performance. I focus on: (i) factors related to model selection and hyperparametrization, to contextualize the choices I make in this work; and (ii) comparisons between models, to contextualize the experiments ran and their results.

### 2.2.1 Hyperparametrization

RS models are often cited for their superior in-sample fit, which is useful for explaining historical phenomena. Dacco; Satchell (1999) noted, however, that even minor errors in predicting the future regime can propagate through the nonlinear structure and lead to worse forecasts than linear alternatives. Standard metrics like mean squared error may also be ill-suited for evaluating nonlinear time series, potentially masking these models’ ability to capture turning points or specific economic states.

A primary challenge in RS modeling is managing the trade-off between flexibility and overfitting. The most critical decision is the number of regimes: too few can underfit, while too many can lead to overparameterization. The regimes’ distribution will be of special importance for this tradeoff. Allowing all parameters to switch can capture complex dynamics and reduce mis-specification, but doing so when unnecessary often dilutes out-of-sample power (Tan; Wu, 2025).

Each model also has its own specificities. For Markov-switching models, the estimation method matters: for example, EM algorithms can balance accuracy and speed in high-dimensional settings (Akbal, 2024). Moreover, the rule used to translate soft posterior probabilities into hard regime labels affects accuracy, and different rules have different properties (Hall; Pagan, 2025). For deterministic

models, the main challenge is variable selection: identifying the correct threshold variable, delay parameter, or nonlinear combination of variables remains difficult in practice.

### 2.2.2 Comparisons between models

Many papers compare RS models across a range of contexts (Clements; Krolzig, 1998), (Bierbrauer; Trück; Weron, 2004), (Pinson *et al.*, 2008), (Janczura; Weron, 2010), (Elias; Wahab; Fang, 2014), (Chen; Bunn, 2014), (Panopoulou; Pantelidis, 2015), (Verne, 2021), (Aydın; Mermi, 2022). No single model is universally superior. The same context can yield a different best-performing model depending on the objective (Akbal, 2024) – e.g. nowcasting, regime identification, portfolio performance, etc.

TAR models are best employed when regime changes are triggered by a single observable variable with rigid boundaries. They have shown strong performance for financial assets such as gold prices and exchange rates, where transitions tend to be fast rather than gradual (Aydın; Mermi, 2022). However, their reliance on observable triggers is a limitation. In contexts such as offshore wind power, where fluctuations reflect complex, non-observable states, TAR models fail to capture the underlying dynamics and can underperform relative to latent-variable models (Pinson *et al.*, 2008).

STAR models are theoretically appropriate for gradual economic adjustment but often face identification challenges. In many financial applications, the estimated smoothness parameter becomes so large that the model effectively collapses into an abrupt threshold model, which makes the smooth specification inefficient (Aydın; Mermi, 2022). STAR models can nonetheless outperform MS in macroeconomic contexts characterized by explosive volatility, such as GDP growth, where one may need to capture the abrupt transitions typical of recession phases (Verne, 2021).

Markov-switching models (MS/MSAR) are well suited to settings where regimes reflect latent, multi-factor drivers (e.g., market sentiment or meteorology) rather than a single observable index. However, their flexible regime framework can make them more sensitive to mis-specification of the number of regimes Janczura; Weron (2010).

### 3 Theoretical framework

In this section, I define the theoretical framework that guides the rest of this work. First, I define the general structure of RS DGPs, aligning all in a common mathematical representation, and relate the concepts of models and metrics to it. An important idea is the separation of the DGP into RGP and SGP.

#### 3.1 The general regime-switching DGP

Let  $y_t \in \mathbb{R}$  denote the series of interest at time  $t \in 1 : T$ ,<sup>2</sup>  $T \in \mathbb{N}$ . Let  $S \in \mathbb{N}$  denote the number of regimes. The *regime variable* is a vector  $r_t \in \mathbb{R}^S$  of ‘weights’ for each regime, indexed by  $r_t^s$ ,  $s \in 1 : S$ .

In this work, I consider only univariate series.

A DGP can be written in terms of a pair: *regime-generating process* (RGP) and *series-generating process* (SGP). This mirrors the separation between the state/system equation and the output/measurement equation in state-space models. They are functions with parameters  $\Theta_r$  and  $\Theta_y$ , respectively, such that:

$$\begin{aligned} r_t &= \text{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r) \\ y_t &= \text{sgp}(y_{1:(t-1)}, r_t, t; \Theta_y) \\ &= \text{sgp}(y_{1:(t-1)}, \text{rgp}(y_{1:(t-1)}, r_{t-1}, t; \Theta_r), t; \Theta_y) \end{aligned} \quad (3.1)$$

Without loss of generality, I restrict the regime weights to be non-negative and sum to one, i.e.,  $r_t \in [0, 1]^S$  and  $\sum_{s=1}^S r_t^s = 1$ . Notably, the number of regimes  $S$  is a parameter in  $\Theta_r$ , and  $\Theta_y$  is a set of parameters for each regime, indexed by  $\Theta_y^s$ . This means that the SGP can be written as:

$$\text{sgp}(y_{1:(t-1)}, r_t, t; \Theta_y) = \sum_{s=1}^S f_{\text{sgp}}(y_{1:(t-1)}, t; \Theta_y^s) \cdot r_t^s \quad (3.2)$$

Note how each regime is weighted by the regime variable  $r_t^s$ . In the simplest case, this weight is binary ( $r_t \in \{0, 1\}^S$ ) – only one regime is ‘on’, and all others are ‘off’. In some models, such as Smooth Transition, the weights can be continuous, and different regimes can be partially ‘on’ at the same time.

In addition,  $\Theta_y$  could encode different functional forms for each regime. As this is uncommon, I refer to  $f_{\text{sgp}}$  as the *SGP functional form*, or simply *SGP*, and to the set of parameters  $\Theta_y$  as the *regime nature* (RN), as they define what changes across regimes and how. Figure 3.1 illustrates this structure.

Figure 3.1: The general RS DGP structure

$$\begin{array}{c} \text{DGP} = \underbrace{\sum_{s=1}^S \text{sgp}(\cdot; \Theta_y^s) \cdot r_t^s}_{\text{functional form}} \quad \& \quad \underbrace{r_t = \text{rgp}(\cdot; \Theta_r)}_{\text{regime nature}} \\ \underbrace{f_{\text{sgp}}}_{\text{functional form}} \quad \& \quad \underbrace{\{\Theta_y^1, \dots, \Theta_y^S\}}_{\text{regime nature}} \end{array}$$

<sup>2</sup>Let  $a : b := \{a, a + 1, \dots, b\}$  for  $a \leq b \in \mathbb{Z}$ , and  $y_{a:b} := \{y_a, \dots, y_b\}$ .



To construct a diverse set of DGPs, I combine different RGP, SGP, and regime natures. One of the challenges of this work is to choose a comprehensible set of these elements, and analyze their differences in a systematic but manageable way.

In the notation above I omitted the error term inside  $f_{\text{sgp}}$ . Many distributions are relevant, including fat-tailed and skewed ones. The parameters  $\Theta_y$  can even encode different error distributions across regimes. By imposing the same distribution across regimes, with the possible exception of a multiplicative factor, the notation and implementation is simplified. I then write the DGP as a function that receives a sequence of random errors and returns the series and regimes:

$$(y_{1:T}, r_{1:T}) = \text{dgp}(\varepsilon_{1:T}; \Theta_r, \Theta_y) \quad (3.3)$$

Consider the notation shorthand  $y := y_{1:T}$ , and similarly for other variables, throughout the rest of this work.

Let the set of considered DGPs be  $P$  (for ‘processes’). These appear in the literature, as discussed in Section 2, and I define them in Section 5.

### 3.2 Models

Consider a model  $\text{mod}$  as a function with (hyper-) parameters  $\Theta_m$  that generates fitted values and  $H$ -step-ahead predictions of the series and regimes. The model can also return a set  $\hat{\pi}$  of metadata, e.g. the estimated coefficients.

$$(\hat{y}, \hat{r}, \hat{\pi}) = \text{mod}(y_{1:(T-H)}; \Theta_m) \quad (3.4)$$

Notably, the number of estimated regimes  $\hat{S}$  is a parameter in  $\Theta_m$ , which may or may not be equal to  $S$ . Let the set of models be  $M$  (for ‘models’). Also present in the literature, they will be defined in Section 5.1.

### 3.3 Regime-conditional metrics

A regime-conditional (RC) metric  $\text{met}$  is a function that takes a series and regimes, and returns a sequence with one value for each regime. I use these metrics to characterize the distribution of  $y_t$  and  $(y_t, y_{t-j})$  within each regime.

$$\text{met} : (y, r) \mapsto \mathbb{R}^S \quad (3.5)$$

An example is the function that returns, for each regime  $s$ , the mean of the series weighted by  $r_t^s$ . This can be done for many common metrics, and is equivalent to mapping  $(y, r)$  to the  $S$  sets  $R_s$  of regimes’ observations,<sup>3</sup> then applying the metric to each set. The benefit of the first approach is that it is more general, allowing for non-binary – i.e., smooth transition – regimes.

For the joint distribution  $(y_t, y_{t-j})$ , the metrics are more complex, as they must consider only the windows  $(y_t, \dots, y_{t-j})$  fully contained in the same regime instance. This is further described in Appendix A.2.

In any case, RC metrics can pool observations from different time windows. To interpret them as regime-specific distributions, I require the series to be stationary within each regime. This requirement restricts the DGPs that this work considers.

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<sup>3</sup> $R_s := \{y_t : r_t^s = \max\{r_t\}\}.$

### 3.3.1 Within-regime stationarity

For the distribution of a regime to be well defined, all datapoints within it must be drawn from the same distribution. Formally, *within-regime strong stationarity* requires, for all regimes  $s \in S$ , a CDF  $F_s$  such that:

$$F_s(y_{t_1}, \dots, y_{t_n}) = F_s(y_{t_1+j}, \dots, y_{t_n+j}) \quad \forall \{y_{t_1}, \dots, y_{t_n}\}, \{y_{t_1+j}, \dots, y_{t_n+j}\} \subset R_s, \quad \forall j \in \mathbb{N}_* \quad (3.6)$$

As I characterize the distributions with specific metrics, weaker assumptions can be made. Restricting attention to moments of the (joint) distribution, I can impose the weak version. Formally, *within-regime weak stationarity* requires,<sup>4</sup> for all  $s \in S$ , that:

$$\begin{aligned} E[y_t] &= E[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ \text{Var}[y_t] &= \text{Var}[y_{t'}] & \forall y_t, y_{t'} \in R_s \\ \text{Cov}[y_t, y_{t-j}] &= \text{Cov}[y_{t'}, y_{t'-j}] & \forall \{y_t, \dots, y_{t-j}\}, \{y_{t'}, \dots, y_{t'-j}\} \subset R_s, \\ & & \forall j \in \mathbb{N}_* \end{aligned} \quad (3.7)$$

Processes with non-binary  $r_t$ , i.e., smooth transitions, do not have truly separated regimes and therefore generally do not satisfy the conditions above. In these cases, the metrics cannot be interpreted as, for example, “the mean of all observations in a regime”. Still, their information may be useful, as I study in this work.

### 3.3.2 Aspects of RC metrics usage

There are two important aspects of the RC metrics usage. First is whether to use the whole sequence of values for each  $s$ , or to condense it into a single value of dispersion across regimes. An example of the latter is the ‘average pairwise distance between the RC means’, a single value that describes how distant the levels of the regimes are. This is equivalent to composing a dispersion function  $\text{disp}$ :

$$\text{disp} \circ \text{met} : (y, r) \mapsto \mathbb{R}^S \mapsto \mathbb{R} \quad (3.8)$$

Second, which series to use: the true or estimated ones. One can use the true values  $(y, r)$  to get the characteristics of the true DGP, and the estimated values  $(\hat{y}, \hat{r})$  or  $(y, \hat{r})$  to get the characteristics of the estimated model.<sup>5</sup> Another option is to calculate the difference between the former and the latter.<sup>6</sup> Another option is to calculate the metric of the difference  $(y - \hat{y}, r)$  or  $(y - \hat{y}, \hat{r})$ .

Less generally, sometimes there are other possible estimators for the same population RC metric, rather than using  $(\hat{y}, \hat{r})$  directly. A special case arises when the metric is a moment of the (joint) distribution and the SGP is simple: one can plug the estimated parameters into the analytical formula for the moment and obtain a better estimator. This is further discussed in Appendix A.2.

These options can be mixed-and-matched, depending on the question of interest. In this work, I use estimated metrics computed from  $(y_{1:(T-H)}, \hat{r}_{1:(T-H)})$ , as these are the values available to the

<sup>4</sup>The ACF condition should be read as “ $j$ ’th autocorrelations between time-points within the same regime instance should always be equal”, not as “ $j$ ’th autocorrelations of every time-point in the same regime instance should always be equal”, although the latter is true for AR processes with order higher than  $j$ .

<sup>5</sup>Note that the value of  $S$  and  $\hat{S}$  can be different, and thus, so the dimension of the metric’s output.

<sup>6</sup>This is only possible if  $S = \hat{S}$  and there is an unambiguous way to match the estimated and true regimes.

econometrician in practice, while I calculate the true metrics analytically. I also ignore the regime-specific metric values and instead condense them by considering only their dispersion, as this is simpler and more comparable across DGPs and models.

Let the set of metrics  $(\text{disp} \circ \text{met})$  be  $C$  (for ‘criteria’). I define these in Section 5.4, but they are mostly based on moments of  $y_t$  and of the pair  $(y_t, y_{t-j})$ ,  $j \in \mathbb{N}$ .

One can also describe the RGP using information such as each regime instance’s average duration, the transition probabilities, and measures derived from them. I use these as control variables in the regression analysis. Finally, regime-unconditional metrics can also be useful.

## 4 Simulation framework

One goal of this work was to define the theoretical framework from the previous section in a general and expandable way, so it can support different exercises, including ones not considered here. I designed the simulation structure with the same goal.

There are the following steps to perform the simulations:

1. Generate random errors for all the DGPs.
2. For each DGP and simulation, generate the series and regimes, producing  $(y, r)$ .
3. For each DGP and simulation, estimate each model, obtaining  $(\hat{y}, \hat{r})$ .
4. For each DGP, simulation, and model, compute each metric.
5. Aggregate the metrics, performance information, and DGP and model descriptors into a dataset.

### 4.1 Hyperparameters

For the forecast performance, I focus on 1-step ahead predictions. It would be interesting to extend this, with or without locally-projected models.

To obtain more than one prediction per simulation, I simulate a  $T - H$ -long series and obtain  $H$  predictions. There are two possible approaches:

1. For each iteration  $h \in 1 : H$ , the model is estimated with the window  $h : (T - H + h - 1)$ , and generates  $\hat{y}_{T-H+h}$ .
2. The model is estimated once with the window  $1 : (T - H)$ , then for each  $h$ ,  $\hat{y}_{T-H+h}$  is generated using  $y_{1:(T-H+h-1)}$ .

The second approach is computationally cheaper, allowing for more simulations and DGPs to be considered. It is the one used in this work, but note that it is less accurate to what would be done in practice, as econometricians often re-estimate their models with new data.

Overall, the hyperparameters of the simulation are as follows:

- Number of simulations:  $I$ . Its main effect is on the precision of the results, and diversity of series.
- Forecast horizon:  $H$  predictions of 1-step ahead values. It also affects the precision of the results, but does not change the diversity of series.
- Total number of observations:  $T$ . Its main effect is on the ability of the models to learn the dynamics and separate the regimes. Results for higher  $T$ 's are more relevant for contexts with a lot of data, such as high-frequency financial data, while lower  $T$ 's are more relevant for contexts with less data, such as macroeconomic data.
- Burn-in period:  $B$ . Its main effect is on reducing dependence on initial values, but with stationary processes this is not too problematic.

Let  $i \in 1 : I$ ,  $I \in \mathbb{N}$  be the simulation index.

## 4.2 Simulation algorithm

I only consider DGPs with the same error distribution – but note that a DGP can still have a volatility parameter multiplying its error. For each DGP, indexed by  $p \in 1 : |P|$ , there are  $I$  random error vectors, each of size  $T$ . Let  $E$  denote the set of all errors. Let  $E_{p,i}$  denote the vector of errors generated for the  $p$ -th DGP and the  $i$ -th simulation. Similar indexing definitions are used for similar collections throughout this document.

Let  $Y$  and  $R$  denote the sets of generated series and regimes. They are computed given  $E_{p,i}$ :

---

```

1: Initialize  $Y$  and  $R$ 
2: for  $p$  in  $1 : |P|$  do
3:   for  $i = 1$  to  $I$  do
4:      $Y_{p,i}, R_{p,i} \leftarrow P_p(E_{p,i})$ 
5:   end for
6: end for

```

---

Now, for each simulation, I estimate each model, generating the sets  $\hat{Y}$ ,  $\hat{R}$ , and  $\hat{\Pi}$ . The models are trained using only  $y_{(B+1):(T-H)}$ , to avoid the burn-in period and leave space for the forecast horizon. The nesting order is the same, but with an additional inner loop for the model estimation.

---

```

1: Initialize  $\hat{Y}$  and  $\hat{R}$ 
2: for  $p$  in  $1 : |P|$  do
3:   for  $i$  in  $1 : I$  do
4:     for  $m$  in  $1 : |M|$  do
5:        $\hat{Y}_{p,i,m}, \hat{R}_{p,i,m}, \hat{\Pi}_{p,i,m} \leftarrow M_m(Y_{p,i}, R_{p,i})$ 
6:     end for
7:   end for
8: end for

```

---

Then, for each model, the dispersion of the RC metrics are calculated and stored as columns of a dataset  $D$ . Each row of  $D$  is identified by  $(p, i, m)$ .

---

```

1: Initialize  $D$ 
2: for  $(p, i, m)$  in  $(1 : |P|) \times (1 : I) \times (1 : |M|)$  do
3:   for  $c$  in  $1 : |C|$  do
4:      $D_{(p,i,m), c} \leftarrow C_c(Y_{p,i,m}, R_{p,i,m}, \hat{Y}_{p,i,m}, \hat{R}_{p,i,m})$ 
5:   end for
6:    $\hat{\Pi}_{p,i,m}$  is appended to  $D_{p,i,m}$ .
7:   Categorical variables  $(p, i, m)$  are appended to  $D_{p,i,m}$ .
8: end for

```

---

Recall the discussion in Section 3.3 about the two different aspects of RC metrics usage. With different options, the function  $C_c$  can use different inputs ( $Y, R, \hat{Y}, \hat{R}$ , or  $\hat{Y}, \hat{R}$ ), which is represented by all four objects being passed to it. Additionally, the function could return the whole sequence of RC metrics, not a single value, then, each row would be identified by  $(p, i, m, s)$ . While the metrics receive the full dataset, estimation is only done with  $(B + 1) : (T - H)$ .

The dataset  $D$  is already in a friendly format for analyzing the relationship between each observation's performance and the regimes' characteristics, and for stratifying by DGP and model.

## 5 Implementation

The framework described in the last two sections is general and accommodates many exercises. In this work, I focus on a specific set of DGPs, models, and metrics. I describe them here.

I focus on regime-switching DGPs with two regimes ( $S = 2$ ). Studying how well models identify regime dynamics with a different number of regimes is an interesting extension. I also include a no-RS model as a baseline.

I choose hyperparameters to balance the DGP ‘population’. I restrict the SGP to a simple stationary  $AR(1)$  process. I consider Markov-switching, Self-Exciting Threshold, and Smooth Transition RGP, each equally represented, with symmetric and asymmetric variants. For each of the three  $AR(1)$  parameters, I consider two RNs: a ‘small’ and a ‘large’ change. I choose the related hyperparameters guided by the concept of ‘regime separation’, described in Section 6. I also include two additional models: K-Means and Random Forest.

Model choice and hyperparametrization are more flexible because they do not affect the experiments’ ‘population’. I use each RGP’s empirical model counterpart with a ‘generic’ hyperparametrization. Increasing model diversity would be a useful extension.

I limit the metrics to essential descriptors of the regime distributions: the first and second moments and the lag-1 autocorrelation. This set could be expanded. I also define performance and RGP-related metrics for the regression analysis.

Finally, I discuss diagnostics for series generation and model estimation and describe the final dataset.

### 5.1 Considered SGPs

The SGP functional form may matter through its interaction with the other ingredients of the DGP. Some applications require specific SGPs, such as conditional volatility in finance and GARCH models. Here, this is not the main point of interest. I therefore consider only an  $AR(1)$  process for its simplicity, popularity, and ease of estimation.

I also assume a Gaussian distribution for the error term, ignoring fat-tailed and skewed alternatives. The distribution is regime-invariant, except for the multiplicative variance parameter  $\sigma$ .

As discussed, I assume within-regime weak stationarity, even though many interesting DGPs are non-stationary. This restricts the absolute value of the  $AR(1)$  parameter to be less than 1. I consider only the following SGP functional form:

$$\begin{aligned} f_{\text{sgp}}(\cdot; (\mu^s, \rho_1^s, \sigma^s)) &= \mu^s + \rho_1^s y_{t-1} + \sigma^s \cdot \varepsilon_t \\ \varepsilon_t &\sim \mathcal{N}(0, 1) \\ |\rho_1^s| &< 1, \quad \sigma^s > 0, \quad \forall s \in 1 : S \end{aligned} \tag{5.1}$$

Several other SGPs could be considered, including transformations of  $y_t$  as regressors, nonlinear regression forms, or decision trees, as in the common Markov-switching Random Forest model. The  $AR(1)$  remains an essential building block, and its simplicity helps isolate the effects of the other ingredients.

## 5.2 Considered RGPs and models

The next ingredient is the RGP. I consider Self-Exciting Threshold, Smooth Transition, and Markov-switching. I also include a no regime-switching benchmark.

Each RGP has an empirical model counterpart, which I also consider.

The formal definition of each RGP/model is presented in Appendix A, first the RGP hypothesis, then the empirical model estimation strategy.

For each RGP, I consider a symmetric case with equally likely regimes and an asymmetric variant.

- **No Regime Switching:**

- Always in regime 1.

- **Self-Exciting Threshold:**

- Fixed hyperparameters: switching based on  $y_{t-1}$ . Different lags often reflect timing choices and are not considered here.
- A single threshold at 0.5, and a single threshold at 0.9.

- **Smooth Transition:**

- Fixed hyperparameters: switching based on  $y_{t-1}$ , the logistic CDF as transition function.
- A single threshold at 0.5, and a single threshold at 0.9.

- **Markov-switching:**

- Symmetric matrix, high persistence ( $P(s|s) = 0.9$ ).
- Asymmetric matrix, high persistence ( $P(1|1) = 0.9$ ,  $P(1|2) = 0.3$ ).

I considered a structural-break model, but analyses suggested it behaved differently from the other models because it does not generate recurring regimes, as expected. I therefore dropped it from the final set of models.

I choose values to target a 50% proportion of regime 1 in the symmetric case and 75% in the asymmetric case for each DGP. The Table 5.1 shows the absolute deviation of the regime 1 proportion from 0.5; I target values close to 0 for the symmetric case, and close to 0.25 for the asymmetric case. RGP and RN interact, so the same RGP can yield different regime proportions across RNs.

Table 5.1: Proportion of regimes across DGPs

|                    | Uncond. | $\mu$ (0, 0.5) | $\mu$ (0, 1) | $\rho_1$ (0.1, 0.9) | $\rho_1$ (0.4, 0.6) | $\sigma$ (1, 1.5) | $\sigma$ (1, 2) |
|--------------------|---------|----------------|--------------|---------------------|---------------------|-------------------|-----------------|
| No RS              | .50     | .50            | .50          | .50                 | .50                 | .50               | .50             |
| MS, $p_{21} = 0.1$ | .02     | .03            | .02          | .03                 | .02                 | .03               | .02             |
| MS, $p_{21} = 0.3$ | .23     | .24            | .23          | .23                 | .23                 | .24               | .23             |
| SET, $\tau = 0.5$  | .12     | .00            | .24          | .12                 | .01                 | .18               | .19             |
| SET, $\tau = 0.9$  | .19     | .18            | .04          | .25                 | .15                 | .28               | .27             |
| ST, $\tau = 0.5$   | .13     | .00            | .13          | .19                 | .25                 | .11               | .08             |
| ST, $\tau = 0.9$   | .19     | .12            | .02          | .30                 | .34                 | .21               | .16             |

Two additional models, with no RGP counterpart, are considered: Random Forest and K-Means clustering. Both receive four lags of the series, and the rolling average, ACF(1), and SD of the series. The RF does not generate a regime series, and is included as benchmark in only some of the analyses. The RF uses 50 trees and 10 minimum terminal node size.

The K-Means is an interesting model because it does not rely on a specific assumption on the RGP structure. The Random Forest is interesting because, while it does not directly model the states, the branching structure of decision trees can capture regime-like dynamics.

For the models, most hyperparameters are as follows:

- I allow all coefficients to change across regimes; this is a common assumption, particularly under possible mis-specification.
- I fix the number of regimes  $\hat{S}$  rather than estimate it. Models are estimated with 2 regimes.
- Model-specific hyperparameters match the related RGP values.

### 5.3 Considered regime natures

The following regime natures are considered, each representing a different way in which the SGP parameters change across regimes:

- **Mean ( $\mu$ ) change:**
  - Small difference:  $(\mu^1 = 0, \mu^2 = 0.5)$
  - Large difference:  $(\mu^1 = 0, \mu^2 = 1)$
- **Persistence ( $\rho_1$ ) change:**
  - Small difference:  $(\rho_1^1 = 0.4, \rho_1^2 = 0.6)$
  - Large difference:  $(\rho_1^1 = 0.2, \rho_1^2 = 0.8)$
- **Volatility ( $\sigma$ ) change:**
  - Small difference:  $(\sigma^1 = 1, \sigma^2 = 1.5)$
  - Large difference:  $(\sigma^1 = 1, \sigma^2 = 2)$

I order regimes by increasing value of the parameter of interest. In the asymmetric RGPs, the rarer regime is always the second one, with the higher value of the relevant parameter.

I choose values to match the regimes' proportion discussed in the last section, and to generate a reasonable level of regime separation, as described in Section 6.

### 5.4 Considered metrics

The goal with RC metrics is to capture changes in the series' characteristics across regimes. One important option is the estimated parameters of the model for each regime, e.g.,  $(\hat{\rho}_s)_{s \in \hat{S}}$ ,  $(\hat{\mu}_s)_{s \in \hat{S}}$ , etc. One might expect these to outshine all other metrics, but in more complex cases where more than one parameter changes, this becomes less useful. More general metrics benefit from their abstraction over the DGP. Additionally, in simple SGPs, there often is a metric that is directly connected to changes in parameters, such as the conditional average for changes in intercept.

In this work, I focus on the moments of the distribution of  $y_t$  and  $(y_t, y_{t-j})$ . Specifically, the RC metrics considered are the RC mean, RC standard deviation, and RC autocorrelation of lag 1. Higher lags could be considered, but in the simple  $AR(1)$  context this would add little additional information.

As stated before, the RC mean and RC SD are simply the mean and SD of each set  $R_s$ . The autocorrelation is similar, but must be calculated separately for each concurrent set of observations in  $R_s$ . The formal definitions are stated in the Appendix A.2.



Because the focus is on the dispersion of RC metrics, two important measures to consider are the standard deviation and the average pairwise absolute difference. For only two regimes, they are very similar and the absolute difference is more intuitive. I define the metrics so that all  $\text{disp} \circ \text{met} \in C$  return a single real value, and  $\text{disp}(x) = |x_1 - x_2|$ .

There are some possible expansions on this work's metrics calculation. One is to use non-standard weights for the empirical moments, giving more importance to observations near the edges of regimes' instances. Another is to use a cluster separation measure, such as the silhouette score, instead of a simple absolute distance between the RC metrics. Finally, one can use distribution distance metrics. These are not currently considered. No regime-unconditional metrics are considered. The list of considered metrics is as below:

- First moment: RC mean  $\hat{\mu}(y|S)$ .
- Second moment: RC standard deviation  $\hat{\sigma}(y|S)$ .
- First autocorrelation: RC lag-1 autocorrelation  $\hat{\rho}_1(y|S)$ .

#### 5.4.1 Performance and RGP metrics

The performance metrics considered are the RMSE for forecasting performance. The MSE is not included, following Dacco; Satchell (1999). The fit performance is measured by  $R^2$  for  $y$  and binary mean error for  $r$ .

I also include RGP-related metrics in the regression analysis: the number of regime switches divided by  $T$ , as a measure of switching frequency; the absolute difference between the average duration of regime 1 instances and regime 2 instances, as a measure of regime asymmetry.

For works with more than two regimes, more complex measures of regime asymmetry can be used, ones that consider the whole matrix of transition probabilities.

### 5.5 Simulation and diagnostics

I implement the simulations and analyze them in the next sections using the R programming language, and the code can be found in [this paper's repository](#). The code is highly modular and fully reproducible, following the intent of setting up an expandable framework.

Following Section 4.1, the chosen hyperparameters are as below. Some values are lower than they could be due to computational constraints. The choice of  $T$  is discussed in Section 6.2.

- Number of simulations:  $I = 500$ .
- Forecast horizon:  $H = 10$  predictions of 1-step ahead values.
- Total number of observations:  $T = 100$ .
- Burn-in period:  $B = 4$ .
- As described above, there are 6 RNs, 7 RGPs, and 4 models.

I generate the error sequences in parallel, using `rTRNG::rnorm_trng`. I estimate the models with `stats::lm`, `mbreaks::dofix`, `tsDyn::setar`, `tsDyn::lstar`, `MSwM::msmFit`, and `randomforest::randomForest`.

Beyond visualizing the series and guaranteeing no missing values, I perform diagnostics on the simulation, model estimation, and metrics calculation. All of the diagnostics are presented in Appendix B.

The errors should be i.i.d. Gaussian with mean 0 and should not present any pattern, especially across the parallelization structure. This is guaranteed by the TRNG library, but I also check it.

Some observations had to be removed. Table 5.2 lists the reasons and the amounts; some observations have multiple issues, which explains the difference between the ‘% Bad’ and ‘% Removed’ columns. Some models did not converge and produced no output. Others could not estimate some parameters. The need to remove those is straightforward.

Some models’ predictions were dominated by one regime and produced zero or only one observation of the other. While this is not a failed estimation, calculating the dispersion of regimes’ distributions is impossible in these cases, and they would be removed from most analyses anyway. Appendix B.3 plots the distribution of regimes’ proportions, and notably, K-Means has the most balanced regimes.

The last removal is less straightforward. Some models generated unreasonably large errors, and parameters unreasonably outside the normal range, so they would commonly be disregarded. This is more subjective and prone to cherry-picking; thus I was parsimonious in this removal. Only RMSEs higher than 50, means higher than the 90th quantile of the whole dataset, and  $\rho_1$ s higher than 10 standard deviations of all the  $\rho_1$  estimates were removed.

Appendix B.2 shows the distribution of fits and RMSEs. Notably, the RF shows signs of overfitting, because the chosen generic hyperparameterization is overkill for the simple SGPs considered. Appendix B.4 shows the distribution of the estimated  $AR$  parameters; ST has distributions with more variance. It also shows distributions of other model parameters. The RF feature importance ranks the first two lags and the rolling average as most important, while the ACF and SD have much lower importance.

Table 5.2: Estimation issues

|                         | Bad obs.     | % Bad       | % Removed   | Obs. left      |
|-------------------------|--------------|-------------|-------------|----------------|
| Full sample             | 0            | 0%          | 0%          | 126,000        |
| Bad convergence         | 148          | 0.1%        | 0.1%        | 125,852        |
| NAs in parameters       | 245          | 0.2%        | 0.2%        | 125,607        |
| Low #obs. in regimes    | 967          | 0.8%        | 0.6%        | 124,859        |
| Unreasonable parameters | 1,803        | 1.4%        | 1.3%        | 123,162        |
| Unreasonable errors     | 145          | 0.1%        | 0%          | 123,137        |
| <b>Total</b>            | <b>3,308</b> | <b>2.6%</b> | <b>2.3%</b> | <b>123,137</b> |

The average coefficient generated by the models is presented in Appendix B.4. Only the matched model-RGP pairs are included, and a test of difference against the true parameter is performed. Many tests don’t pass, but that is expected, as all parameters are allowed to change across regimes. It is important to note that they generally do (except for  $\sigma$ ).

To consider the metrics estimation, the regime-conditional and unconditional moments are estimated using and tested against their true values in Appendix B.5. The estimations use  $(y, r)$ , while the true values are calculated via the analytical formula (the unconditional true values are not calculated). Again, the tests are not generally expected to pass, especially given their high power.

As a final placebo test, the forecast performance (RMSE) was regressed against the index  $i$  of the simulation. The Table B.1 shows generally no relation, as expected.

## 6 Regimes' distribution

In this section, I explore what the regimes' distributions, as captured by the RC metrics, reveal about regime DGPs and models. This helps interpret the systematic results of the next section and provides stylized facts for the econometrician.

The first two subsections, 6.1 and 6.2, summarize, for each DGP, regime separation under each metric, first for the full sample and then as sample size varies. Section 6.3 examines how the models capture that separation.

### 6.1 Regime separation in $T$

I first ask what each DGP implies for the distribution of  $y_t$  across regimes. One could plot regime-conditional distributions (and, ideally, the joint distribution of  $(y_t, y_{t-1})$ ), but these visualizations become difficult to interpret when repeated across many DGPs and hyperparameter settings.

The approach in this work is instead to characterize each regime distribution with regime-conditional metrics, and to summarize “how different the regimes are” by a dispersion across regimes. As stated before, more metrics than the ones considered here could be necessary to fully capture the differences between regimes.

Consider Table 6.1. It should be read as a compact ‘profile’ of the DGP in terms of regime separation. Each *row* corresponds to one DGP configuration, grouped by the regime-generating process (RGP) and the regime-nature parameter (RN); for each RC metric (mean, lag-1 autocorrelation, and standard deviation), there are two *columns* corresponding to the RN’s ‘small’ and ‘big’ parameter changes; each *cell* reports the absolute difference in the corresponding RC metric across the two regimes, as well as the ‘(SD)’ and non-zero-test p-value stars. The table uses only symmetric RGPs, as well as 100 of the simulation indices. The metrics are calculated with  $(y, r)$ .

Table 6.1: Regimes' metrics separation across DGPs

| RGP | RN       | $d(\hat{\mu}(.))$ |               | $d(\hat{\rho}_1(.))$ |              | $d(\hat{\sigma}(.))$ |              |
|-----|----------|-------------------|---------------|----------------------|--------------|----------------------|--------------|
|     |          | (0, 0.5)          | (0, 1)        | (0.1, 0.9)           | (0.4, 0.6)   | (1, 1.5)             | (1, 2)       |
| MS  | $\mu$    | .85*** (.02)      | 1.62*** (.03) | .01 (.01)            | .01 (.01)    | .02 (0)              | .02 (.01)    |
|     | $\rho_1$ | .03 (.06)         | .07 (.3)      | .11*** (0)           | .38*** (.05) | .12*** (.01)         | .53*** (.15) |
|     | $\sigma$ | .03 (.1)          | .01 (.15)     | .05** (0)            | 0 (.03)      | .48*** (.06)         | .99*** (.14) |
| SET | $\mu$    | 1.52*** (.04)     | 2.29*** (.1)  | 0 (.01)              | .22*** (.05) | .01 (0)              | .05*** (.03) |
|     | $\rho_1$ | 1.02*** (.05)     | 1.3*** (.19)  | .02 (.03)            | .22*** (.02) | .02 (.06)            | .22*** (.06) |
|     | $\sigma$ | 1.09*** (.17)     | 1.24*** (.3)  | .13*** (.05)         | .14*** (.06) | .49*** (.1)          | .98*** (.16) |
| ST  | $\mu$    | .64*** (0)        | .47*** (.03)  | 0 (.01)              | .08*** (.03) | 0 (.01)              | .03** (.01)  |
|     | $\rho_1$ | .92*** (0)        | 1.06*** (.07) | .21*** (.02)         | .41*** (.05) | .07*** (.01)         | .22*** (.02) |
|     | $\sigma$ | 1.15*** (.01)     | 1.41*** (.07) | .08*** (0)           | .05*** (0)   | .28*** (.01)         | .58*** (.07) |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

In the first row (MS with a  $\mu$  change), only the mean separates the regimes, as expected. With a change in  $\rho_1$ , both the ACF and the SD differentiate the regimes, because the standard deviation depends on  $\rho_1$ . When  $\sigma$  changes, only SD differentiates the regimes. This suggests a mapping from which metric differentiates the regimes to which parameter changes in the DGP, but this mapping is not universal, because it changes as we consider other RGPs.

MS has the cleanest result because its RGP does not depend directly on  $y$ , so it interacts less with the RN. For (SET,  $\mu$ ), the mean is separated, but the (big) ACF and SD also separate. This happens because in the ‘big’ regime, a higher mean creates a feedback loop that keeps values higher. With a  $\rho_1$  change, the mean also increases: in the ‘big’ regime, higher  $\rho_1$  amplifies large values, raising the mean. For  $\sigma$ , a similar effect occurs: the ‘big’ regime has larger variation and larger maxima, while the ‘small’ regime has smaller variation and smaller minima, increasing the mean difference across regimes.

With ST, we obtain a similar result, as its RGP is very similar to SET. Notably, ST shows higher separation in the ‘small’ RNs, which SET does not show as clearly. Overall, this table exemplifies the importance of the RGP–RN interaction. This is relevant for understanding models’ performance and how metrics’ profiles vary with it.

## 6.2 Regime separation across $t$

It is intuitive that sample size affects whether estimated metrics detect separation across regimes. The ability to learn regime dynamics depends not only on which parameter changes (RN), but also on how quickly the induced regime separation becomes statistically visible as  $T$  increases. To study the interaction between RGP, RN and  $T$ , I calculate each metric with the data up to each time point ( $1 : 2, 1 : 3, \dots, 1 : T$ ). By graphing the last time point considered on the x-axis and the across-regime difference of the RC metric on the y-axis, we can see how separation evolves with sample size.

The figures 6.1 - 6.3 present the results. The x-axis is the last time point included in the ‘effective sample’, used to calculate the line and ribbon at that value; each line is the average (across simulations) of the metrics’ dispersion, and the ribbon is calculated as the standard error (across simulations) times 1.96. Both are stratified by RGP symmetry, indicated via color. The graphs use only the big RNs, and 20 simulation indices. The metrics are calculated with  $(y, r)$ .

Two main results emerge. First, asymmetry mainly affects uncertainty: average values across asymmetric and symmetric cases are very similar, but the standard deviation is not. This is because the less frequent regime contributes few observations, and this number grows very slowly across time.

Second, the values and their standard deviations seem to be more or less stabilized around  $T = 60$ , which means that for a series of this size (60 to 100 observations) we would likely see similar usefulness of these metrics. For smaller series, the metrics are substantially less informative.

## 6.3 Models and regimes

To study how the models’ estimates relate to regime separation, I compare the estimated metrics with the true ones. I do this in Figure 6.4, which shows the absolute difference between the true and estimated dispersion of the RC metrics. Each panel row corresponds to a parameter change, and each panel column to a metric. The estimated metrics are calculated with  $(y, \hat{r})$ , while the true metrics are calculated via the analytical formula; thus, this difference combines metric-estimation error and  $\hat{r}$  error. The metrics were scaled by their median absolute difference (MAD) for comparability.

We can see that KM and ST do well with the mean when  $\mu$  is changing, but not otherwise, while SET has a more balanced result, and MS performs worst. KM overall shines with  $\mu$  changes, possibly because it includes four lags of the series level in its composition. More importantly, the distribution for any given metric varies widely across RNs. This means that no model makes a ‘fixed’ error for each metric: depending on the unobservable RN, the correctness of the estimated metrics varies. In contrast, a graph stratified by RGP would show that each metric error is somewhat invariant across RGPs. This will be relevant in Section 7.2. The breaks in the distributions come from the fact that the true metrics are set values for each DGP.

Figure 6.1: Regime separation - MS

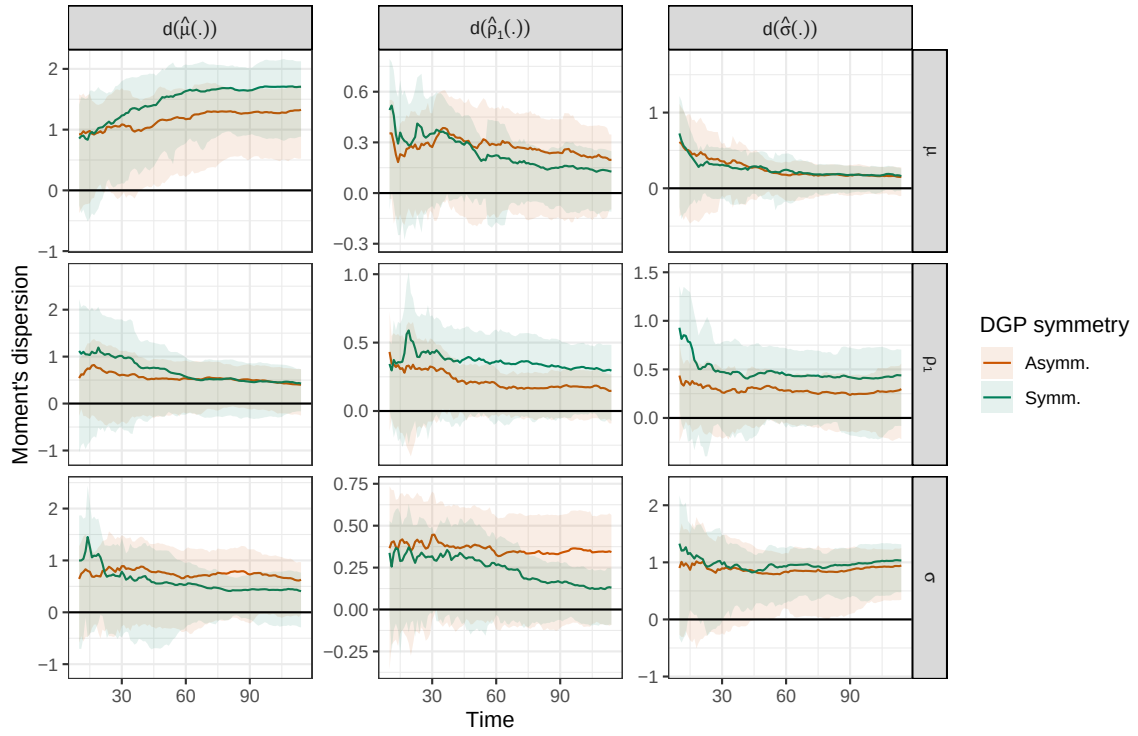


Figure 6.2: Regime separation - SET

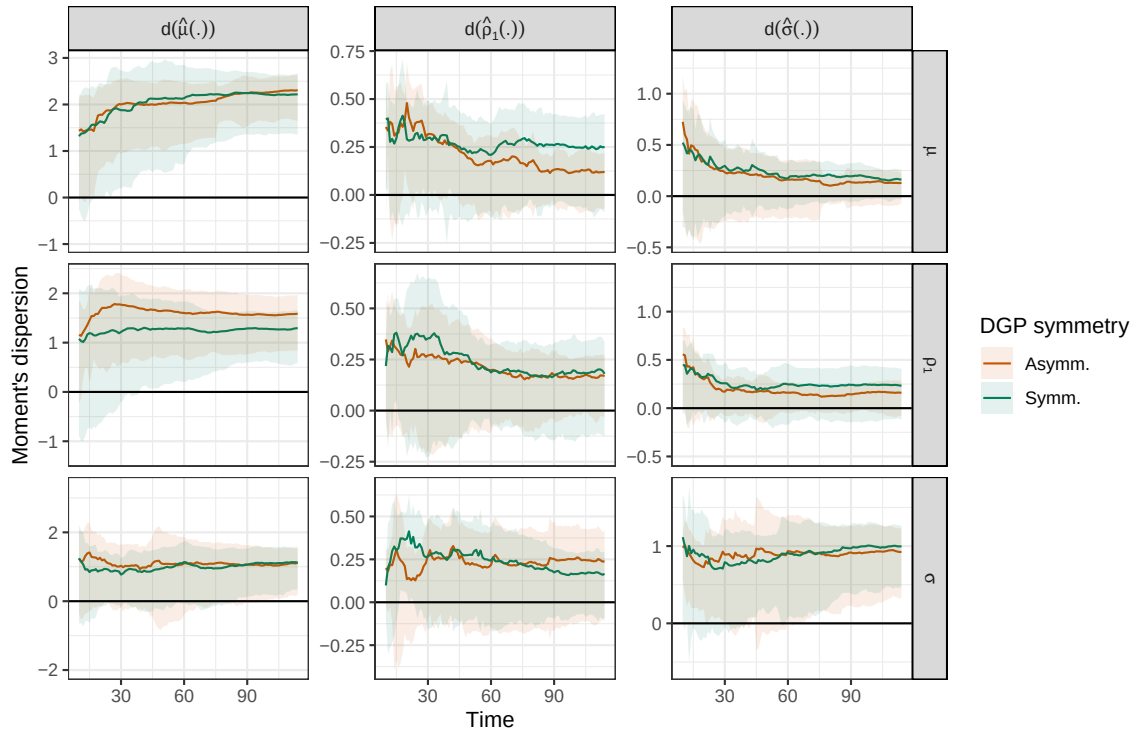


Figure 6.3: Regime separation - ST

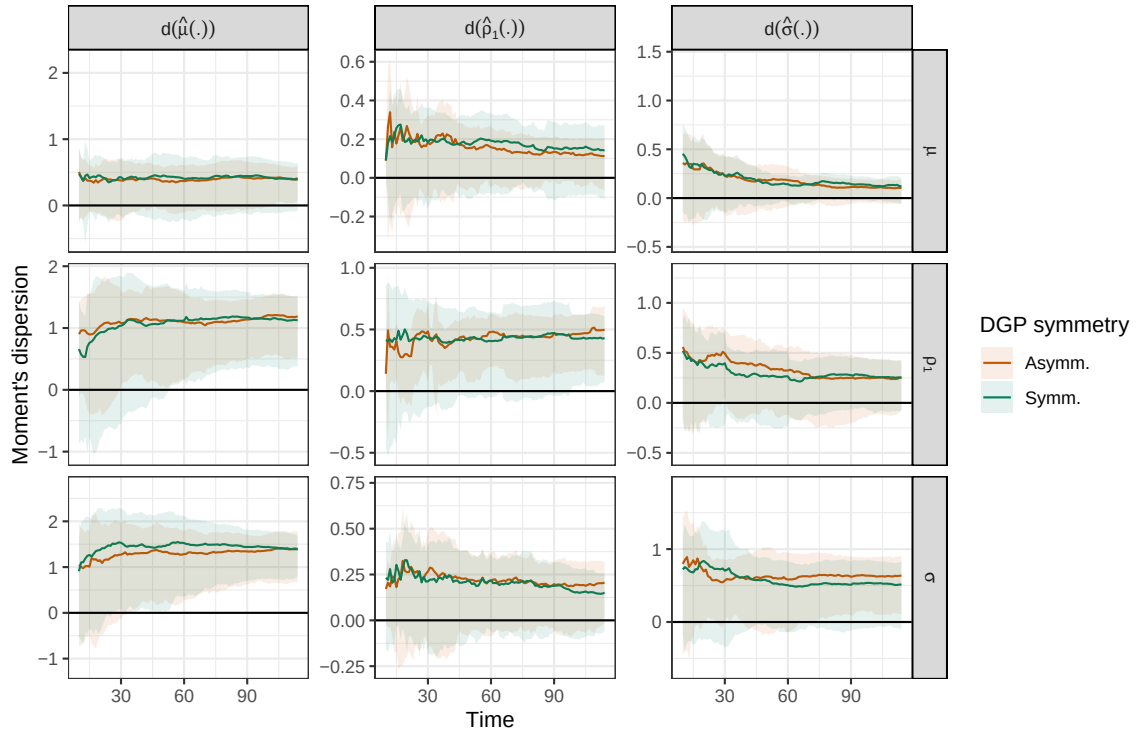
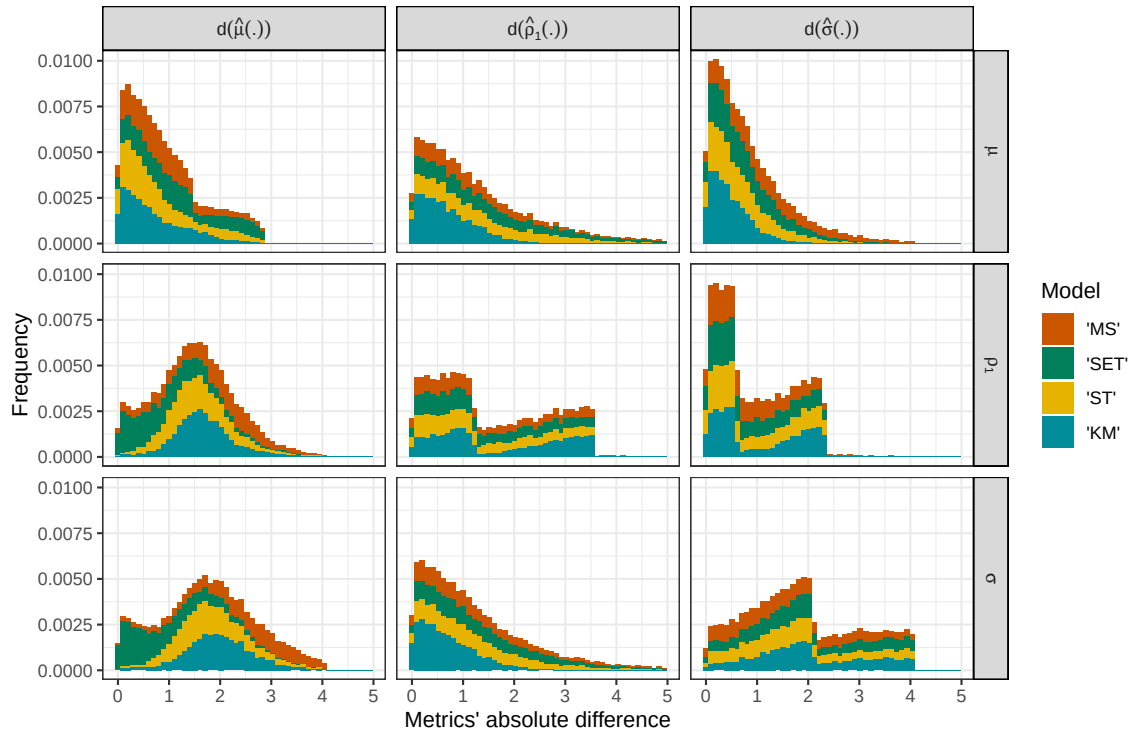


Figure 6.4: Metrics difference



Another useful analysis relates model outputs (estimated fit,  $r$ , parameters) to model performance. The fit and parameters, continuous variables, are studied in Section 7.3. The binarized regime assignment has special importance, as it is directly related to regime separation, and can be used to split the forecasting errors into “correctly identified” and “incorrectly identified” observations. The figures 6.5 - 6.8 compare the distribution of individual forecasting errors, conditional on whether the underlying regime was correctly identified. Each panel corresponds to a DGP configuration and presents the distribution of forecasting errors stratified by regime correctness; each figure corresponds to one model. The graphs are done with big RNs, symmetric RGPs, and 100 simulation indices.

In general terms, the ‘correct’ distributions have slimmer tails, which is in line with the literature. This is not always the case, however, especially for the MS model; this result will be discussed in Section 7.1. For the ST model with no-RS RGP, we have high errors overall and bimodal distributions, which will be discussed in Section 7.2.

Figure 6.5: Regime and series prediction - no-RS

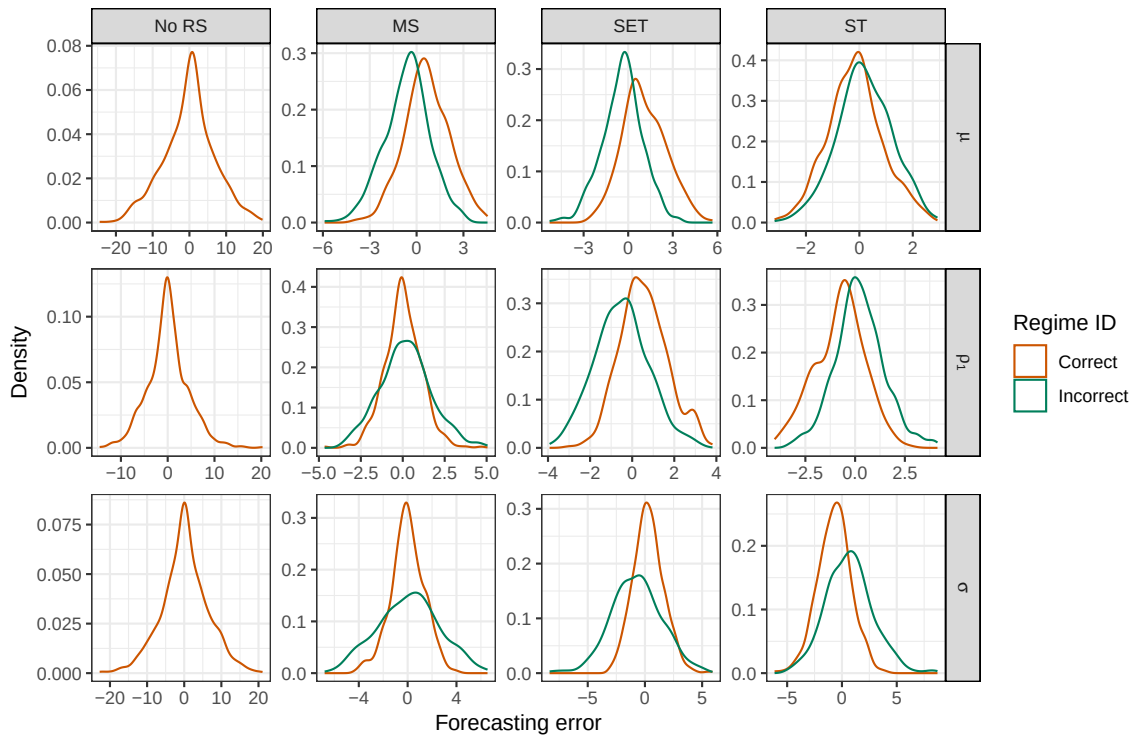


Figure 6.6: Regime and series prediction - MS

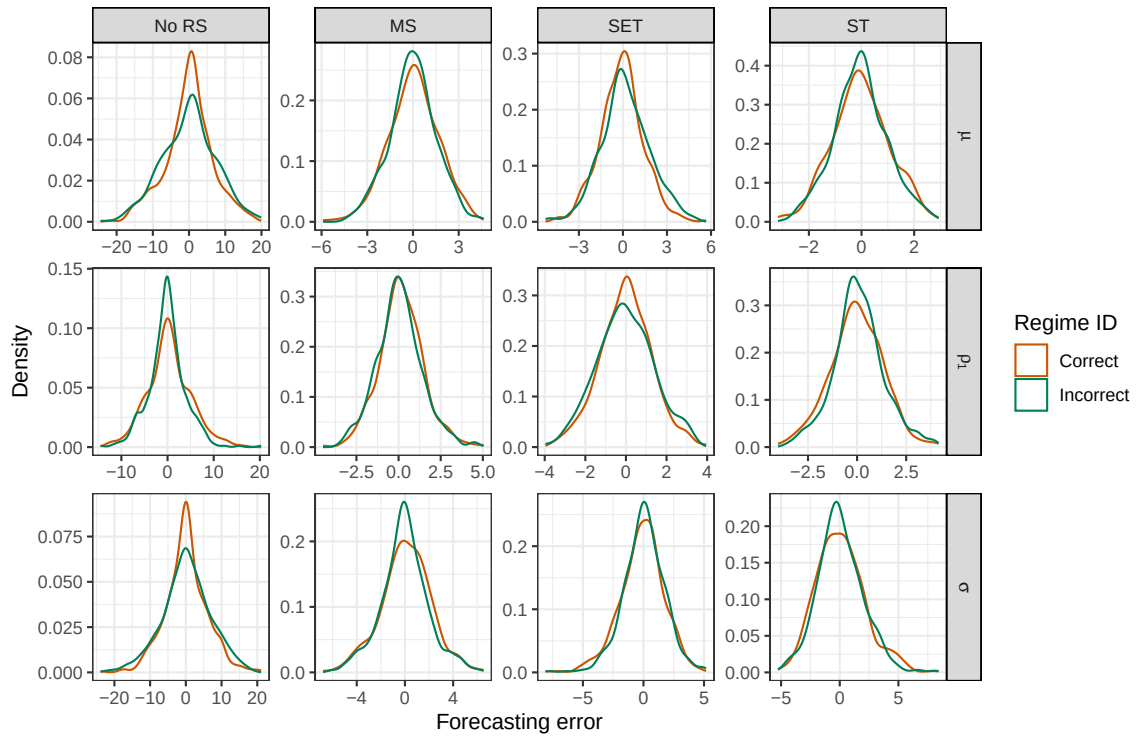


Figure 6.7: Regime and series prediction - SET

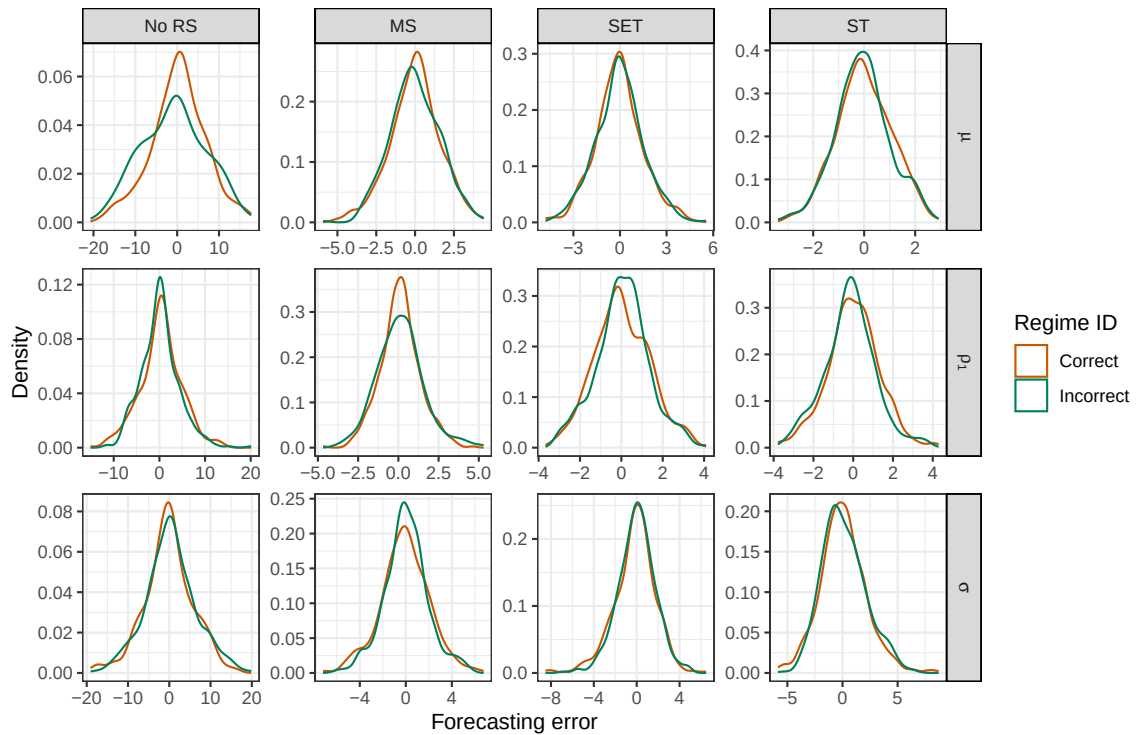




Figure 6.8: Regime and series prediction - ST

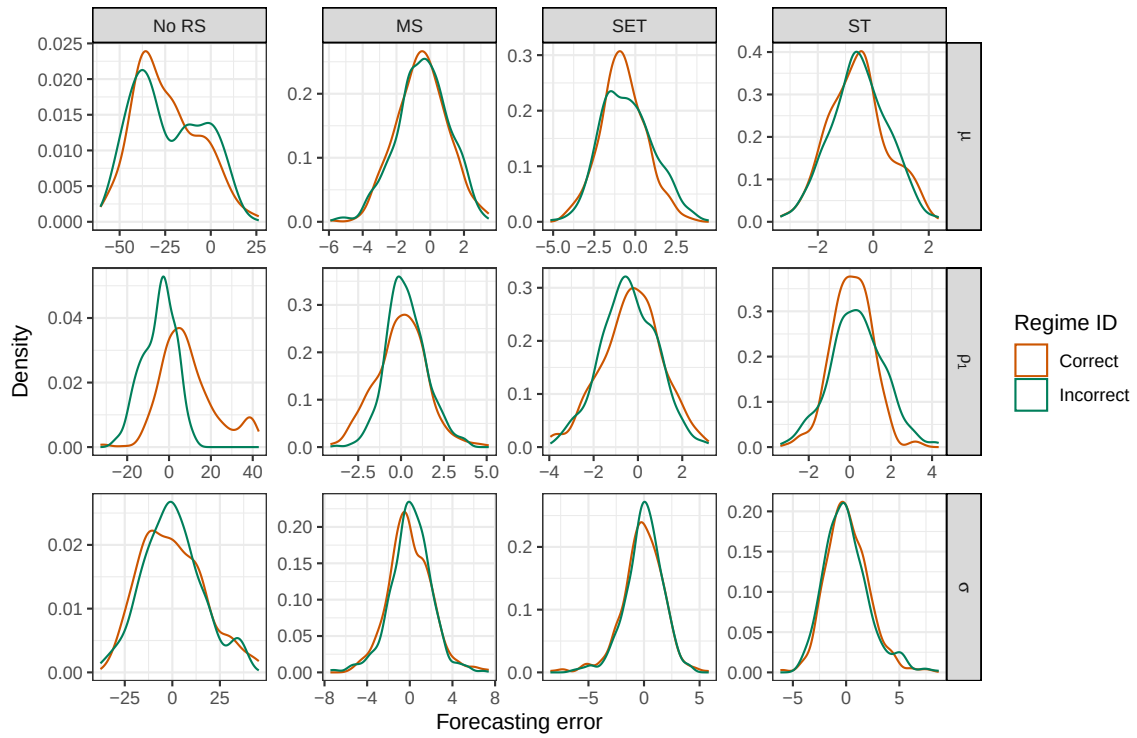
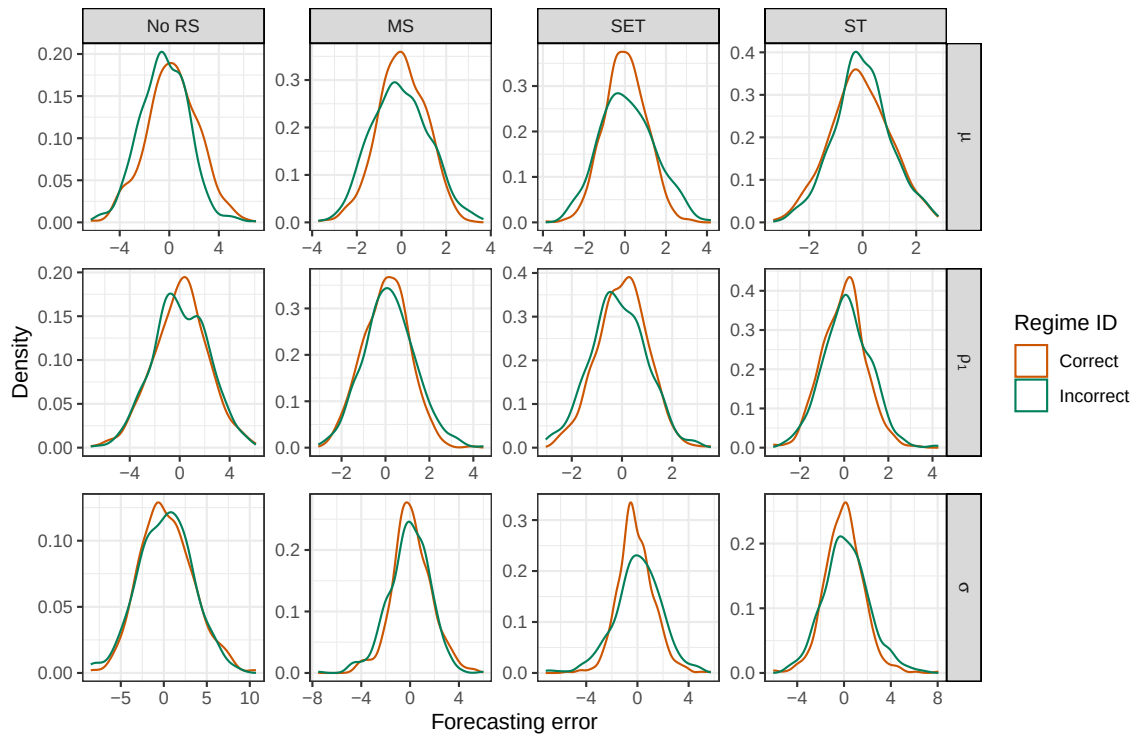


Figure 6.9: Regime and series prediction - KM



## 7 Performance analysis

In this section, the goal is to systematically analyze the performance of models and how the regimes' distributions might affect and inform performance. First, I examine the overall performance of each model through their fixed effects. I (i) add controls to identify which component of the models is most related to their performance; and (ii) stratify the DGPs to understand how performance varies across scenarios.

I analyze the models' interaction with the DGPs. I (i) evaluate the effect of model-RGP mismatch and interact it with the DGP options to determine if the mismatch effect varies across scenarios; (ii) analyze specific model-DGP pairs; and (iii) consider an alternative way to describe the DGPs, in terms of the RC metrics they generate, and assess how each model performs against each profile of series. I consider whether any results suggest practical recommendations for the econometrician.

Then, I analyze which model component (estimated fit,  $r$ , parameters, or metrics) is most associated with each model's performance. For each model, there are different points of interest for the econometrician.

Finally, regarding mis-specification, I test the hypothesis that underestimating the number of regimes is less harmful when the regimes' distributions are less separated, and whether the opposite is true for overestimation.

All regressions use RMSE as the dependent variable, so higher coefficient values imply worse performance associated with the given variable. The metrics and parameters are normalized as  $|x - \text{median}(x)|/\text{mad}(x)$ , except for the RMSE. Some metrics are not available for all observations, such as the ACF, which requires at least one length-2 instance of each regime in the series. Thus, the number of observations in each regression can vary. A  $\Delta$  symbol represents the absolute difference between the estimated value (often with  $y_{1:(T-H)}, \hat{r}_{1:(T-H)}$ ) and the analytical true value.

The results are conditional on the population of DGPs considered and should not be interpreted as universal truths for any series.

### 7.1 Models' fixed effects

The fixed effects are presented in Table 7.1. This excludes the no-RS and RF models. The first column, without controls, indicates that SET and ST perform similarly and better than MS. This may be due to the higher prevalence of threshold-based DGPs compared to Markov-based ones in the DGP pool. All of them perform significantly worse than KM.

To understand how the qualities of the models contribute to these fixed effects, I add controls for matching the true vs. estimated fit,  $r$ ,<sup>7</sup> parameters, and metrics' dispersion, then remove them one by one. A better FE without a control means that the relationship between the model and that control is a positive component in its FE. Comparing (2) and (3), MS shows the highest improvement, given its flexible structure that can fit very general data, compared to the rigidity of the threshold-based models.

With Columns (2) and (4), we observe the counterpart effect: MS struggles to match the regimes, given its geometric distribution that is often not aligned with threshold-based regimes. ST performs slightly worse than SET, but this is because the regimes measure is binarized and does not utilize the continuity of ST regimes. The only model with a positive relationship is KM, which assumes no specific hypothesis on the regime structure. This particular relationship between KM and regime matching will help explain further results.

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<sup>7</sup>Controls include the binarized mean error, the  $\Delta$  average switches, and the  $\Delta$  average duration.

Comparing (2) and (5), none of the models' performance improves by matching the parameters, especially ST, which can often generate very unreasonable estimates, as noted in Appendix B.4. In contrast, matching the metrics explains a large part of the models' fixed effects, especially for ST, but not for KM. This is an important realization, as the effects of parameters and metrics might initially seem interchangeable, but they are not.

Table 7.1: Models' fixed effects - baseline and controls

|                     | RMSE                |                     |                     |                     |                     |                     |
|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                     | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| MS                  | 1.781***<br>(0.009) | 1.226***<br>(0.005) | 1.184***<br>(0.005) | 1.492***<br>(0.008) | 1.254***<br>(0.005) | 1.244***<br>(0.005) |
| SET                 | 1.687***<br>(0.009) | 1.271***<br>(0.005) | 1.251***<br>(0.005) | 1.327***<br>(0.008) | 1.324***<br>(0.005) | 1.250***<br>(0.004) |
| ST                  | 1.632***<br>(0.009) | 1.219***<br>(0.005) | 1.198***<br>(0.005) | 1.370***<br>(0.009) | 1.286***<br>(0.005) | 1.197***<br>(0.005) |
| KM                  | 1.308***<br>(0.009) | 1.068***<br>(0.004) | 1.047***<br>(0.004) | 0.984***<br>(0.007) | 1.096***<br>(0.004) | 1.067***<br>(0.004) |
| $R^2$               |                     | yes                 |                     | yes                 | yes                 | yes                 |
| regimes             |                     | yes                 | yes                 |                     | yes                 | yes                 |
| $\Delta$ params.    |                     | yes                 | yes                 | yes                 |                     | yes                 |
| $\Delta$ metrics    |                     | yes                 | yes                 | yes                 | yes                 |                     |
| Observations        | 79,934              | 62,013              | 62,013              | 68,124              | 62,016              | 71,054              |
| $R^2$               | 0.614               | 0.886               | 0.884               | 0.795               | 0.880               | 0.876               |
| Adjusted $R^2$      | 0.614               | 0.886               | 0.884               | 0.795               | 0.880               | 0.876               |
| Residual Std. Error | 1.277               | 0.466               | 0.469               | 0.878               | 0.477               | 0.492               |

Note:

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01

To understand how these fixed effects vary across different DGPs, Table 7.2 shows the fixed effects with no controls but with stratifications in the dataset. It excludes the no-RS model. Column (1) is the same as before but now includes the RF model, which is second to KM, partially due to its 'generic' overfitted hyperparameterization. While the MS model showed a poorer relationship with regime matching, in column (2), we see that its flexibility pays off, as it performs better in asymmetric regimes. The RF model shows similar behavior. In Column (3), ST deals better with small regime natures than SET. This is related to Table 6.1, where SET series were not fully separated in small regime natures, while in the ST ones they were, and this is reflected in the fixed effects.

Changes in  $\mu$  (4) generate a better setup for separating the series via threshold, improving SET and ST. Changes in  $\rho$  (5) are poorly captured by ST but nicely captured by MS and KM. In a  $\sigma$  change (6), everyone performs worse, as it is overall harder to forecast. The overfitting of RF is evident here, as it presents the highest change from column (1), while ST shows one of the best robustness, in line with its ability to handle explosive volatility, as noted by Verne (2021).

The KM model's FE changes the least across stratifications, on average 0.17, followed by ST, with 0.23. These are characteristics of more general models that have some 'flavor' of universal approximation power. The ability of ST to deal with some of the harder DGPs is related to its relationship to the complexity of a Neural Network, as discussed by Medeiros (2000), granted that the present setup with  $S = 2$  does not reach its full potential. The RF model is also kept shy of its potential, and while it would be interesting to have a better-architected RF, this benchmark already shows its improvement over the traditional models while making it clear that overfit is a risk.

Table 7.2: Models fixed effects - across stratifications

|                         | RMSE                |                     |                     |                     |                     |                     |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                         | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 |
| MS                      | 1.781***<br>(0.009) | 1.308***<br>(0.005) | 1.676***<br>(0.013) | 1.782***<br>(0.015) | 1.569***<br>(0.015) | 1.987***<br>(0.017) |
| SET                     | 1.687***<br>(0.009) | 1.322***<br>(0.005) | 1.607***<br>(0.013) | 1.601***<br>(0.015) | 1.529***<br>(0.015) | 1.932***<br>(0.017) |
| ST                      | 1.632***<br>(0.009) | 1.313***<br>(0.005) | 1.553***<br>(0.013) | 1.391***<br>(0.016) | 1.611***<br>(0.015) | 1.884***<br>(0.018) |
| KM                      | 1.308***<br>(0.009) | 1.132***<br>(0.005) | 1.254***<br>(0.013) | 1.182***<br>(0.015) | 1.180***<br>(0.015) | 1.562***<br>(0.017) |
| RF                      | 1.588***<br>(0.009) | 1.214***<br>(0.005) | 1.578***<br>(0.013) | 1.320***<br>(0.016) | 1.508***<br>(0.015) | 1.910***<br>(0.017) |
| Strat.                  | none                | asym. RGP           | small RN            | $\mu$ RN            | $\rho_1$ RN         | $\sigma$ RN         |
| Observations            | 99,944              | 44,491              | 50,172              | 32,357              | 33,925              | 33,662              |
| R <sup>2</sup>          | 0.603               | 0.879               | 0.591               | 0.589               | 0.595               | 0.639               |
| Adjusted R <sup>2</sup> | 0.603               | 0.879               | 0.591               | 0.589               | 0.595               | 0.639               |
| Residual Std. Error     | 1.302               | 0.467               | 1.280               | 1.229               | 1.224               | 1.398               |

Note:

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01

## 7.2 Model and DGP interaction

### 7.2.1 Mis-specification of RGP family

To study the effect of mis-specification, Table 7.3 defines mis-specification as “the family of the model being different than the family of the RGP”. Only the MS, SET, and ST models are included. The baseline effect of mis-specification is an RMSE increase of 0.52. To further understand how this effect changes across stratifications, consider columns (2-6). (2): MS models suffer more, in line with previous results, while ST models suffer less. (3): Mis-estimating an RS model in a non-RS RGP is disastrous, and in light of that, mis-specification across RS RGPs is not as relevant. This will be further explored in the next subsection.

In (4), asymmetric regimes are harder to estimate, so sometimes mis-specification helps. (5): Similar to before,  $\sigma$  changes make forecasting harder, and this effect is compounded by mis-specification; for the other two parameters, the effect is similar. Finally, (6) shows that in small RNs, estimating the wrong regime generates a smaller error, so mis-specifying the RGP is not too problematic.

### 7.2.2 Specific model-RGP mis-specifications

Misspecification in general is detrimental. However, stratifying by each of the RGP families in Table 7.4 shows that each model (line) has similar values across the different RGPs (columns). This confirms that correctly specifying the RGP family is not the most critical factor for performance.

With SET, the values are positive, meaning that the model performs slightly better with no-RS series. For ST, the values are highly negative, meaning that no-RS series generate disastrous results, and this is reflected in Table 7.3 column (3). Conversely, KM performs exceptionally well when there is no regime switching. For MS, RS or no-RS matters less. The invariance to RGP, especially between

Table 7.3: Models and DGPs mis-specification

|                         | RMSE                |                     |                     |                      |                     |                     |
|-------------------------|---------------------|---------------------|---------------------|----------------------|---------------------|---------------------|
|                         | (1)                 | (2)                 | (3)                 | (4)                  | (5)                 | (6)                 |
| Baseline                | 0.515***<br>(0.013) |                     |                     |                      |                     |                     |
| Model: MS               |                     | 0.627***<br>(0.016) |                     |                      |                     |                     |
| Model: SET              |                     | 0.488***<br>(0.016) |                     |                      |                     |                     |
| Model: ST               |                     | 0.420***<br>(0.016) |                     |                      |                     |                     |
| RGP: No RS              |                     |                     | 3.531***<br>(0.014) |                      |                     |                     |
| RGP: MS                 |                     |                     | −0.016<br>(0.011)   |                      |                     |                     |
| RGP: SET                |                     |                     | 0.006<br>(0.011)    |                      |                     |                     |
| RGP: ST                 |                     |                     | −0.002<br>(0.011)   |                      |                     |                     |
| RGP: asym.              |                     |                     |                     | −0.028***<br>(0.006) |                     |                     |
| RGP: sym.               |                     |                     |                     | 0.020***<br>(0.006)  |                     |                     |
| RN: $\mu$               |                     |                     |                     |                      | 0.400***<br>(0.016) |                     |
| RN: $\rho_1$            |                     |                     |                     |                      | 0.381***<br>(0.016) |                     |
| RN: $\sigma$            |                     |                     |                     |                      | 0.760***<br>(0.016) |                     |
| RN: small               |                     |                     |                     |                      |                     | 0.424***<br>(0.015) |
| RN: big                 |                     |                     |                     |                      |                     | 0.607***<br>(0.015) |
| Constant                | 1.340***<br>(0.011) | 1.340***<br>(0.011) | 1.340***<br>(0.007) | 1.340***<br>(0.004)  | 1.340***<br>(0.011) | 1.340***<br>(0.011) |
| Observations            | 59,092              | 59,092              | 59,092              | 53,009               | 59,092              | 59,092              |
| R <sup>2</sup>          | 0.027               | 0.029               | 0.554               | 0.001                | 0.037               | 0.029               |
| Adjusted R <sup>2</sup> | 0.027               | 0.029               | 0.554               | 0.001                | 0.037               | 0.029               |
| Residual Std. Error     | 1.424               | 1.422               | 0.964               | 0.538                | 1.417               | 1.422               |
| F Statistic p-value     | <0.001              | <0.001              | <0.001              | <0.001               | <0.001              | <0.001              |

Note:

\*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01

SET and ST, is intriguing and, in some ways, contradicts the results of Aydın; Mermi (2022), although a broader diversity of SET and ST models would be required for a proper comparison.

Table 7.4: Models and RGP pairs mis-specification

| Model                      | RGP               |                   |                   |
|----------------------------|-------------------|-------------------|-------------------|
|                            | MS                | SET               | ST                |
| MS                         | 0.001 (0.029)     | 0.005 (0.029)     | 0.007 (0.029)     |
| SET                        | 0.066** (0.03)    | 0.065** (0.03)    | 0.066** (0.03)    |
| ST                         | -1.657*** (0.035) | -1.644*** (0.035) | -1.633*** (0.035) |
| KM                         | 2.113*** (0.028)  | 2.05*** (0.028)   | 2.142*** (0.028)  |
| Observations               |                   | 100840            |                   |
| $R^2$ , Adjusted $R^2$     |                   | 0.826, 0.826      |                   |
| Resid. SE, F stat. p-value |                   | 0.884, <0.001     |                   |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

RGP alone does not explain the mis-specification effect, but the interaction with RN was deemed relevant in the exploratory analysis. Table 7.5 shows the coefficients for the interactions between model, RGP, and RN,<sup>8</sup> but we still observe a similar result of ‘invariance’ within each ‘panel’ (RGP-model pair) of the table, except for KM.

Table 7.5: Models, RGPs, and RNs mis-specification

| RGP                        | SGP      | Model          |                |                  |                   |
|----------------------------|----------|----------------|----------------|------------------|-------------------|
|                            |          | MS             | SET            | ST               | KM                |
| No RS                      | $\mu$    | 0.02 (0.048)   | -0.055 (0.052) | 2.085*** (0.091) | -2.868*** (0.047) |
|                            | $\rho_1$ | -0.015 (0.047) | -0.034 (0.048) | 1.927*** (0.052) | -1.522*** (0.047) |
|                            | $\sigma$ | -0.031 (0.048) | 0.048 (0.05)   | 1.825*** (0.056) |                   |
| MS                         | $\mu$    | -0.012 (0.038) | -0.008 (0.038) | 0.012 (0.038)    | -0.028 (0.038)    |
|                            | $\rho_1$ | -0.011 (0.038) | -0.004 (0.038) | -0.007 (0.038)   | 0.073* (0.038)    |
|                            | $\sigma$ | -0.003 (0.038) | -0.001 (0.038) | -0.004 (0.038)   |                   |
| SET                        | $\mu$    | 0.001 (0.038)  | -0.01 (0.038)  | 0.043 (0.038)    | -0.16*** (0.038)  |
|                            | $\rho_1$ | -0.011 (0.038) | -0.011 (0.038) | 0.008 (0.038)    | 0.003 (0.038)     |
|                            | $\sigma$ | -0.003 (0.038) | 0.006 (0.038)  | -0.017 (0.038)   |                   |
| ST                         | $\mu$    | -0.01 (0.038)  | 0 (0.038)      | 0.082** (0.038)  | 0.175*** (0.038)  |
|                            | $\rho_1$ | -0.006 (0.038) | -0.013 (0.038) | -0.013 (0.038)   | -0.021 (0.038)    |
| Observations               |          | 100840         |                |                  |                   |
| $R^2$ , Adjusted $R^2$     |          | 0.84, 0.84     |                |                  |                   |
| Resid. SE, F stat. p-value |          | 0.848, <0.001  |                |                  |                   |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

<sup>8</sup>Controls include model, parameter change, and RGP family FEs, plus interactions between the last two.

### 7.2.3 Models and RC metrics

Fortunately, the point of this work is that the RC metrics provide another way to characterize the DGP, and in some ways, a more general one than the RGP family ‘label’. I construct similar tables interacting each model with the profiles of (dispersion of) metrics in the series. As the whole profile of metrics is important, all of the interactions of  $\text{mod} \cdot d(\mu(.)) \cdot d(\rho_1(.)) \cdot d(\sigma(.))$  are included as controls.

Table 7.6 uses the true values of the metrics. After controlling for the interactions, SET, ST, and KM have a negative effect when estimated on series with high average separation, given the easier time separating series via thresholds. For MS, we do not observe significant effects, and not much can be said about it without deeper analysis. ST seems to struggle with high separation in ACF. KM benefits from SD differentiation.

Table 7.6: Models and true RC metrics

| Model                      | Metric            |                      |                      |
|----------------------------|-------------------|----------------------|----------------------|
|                            | $d(\hat{\mu}(.))$ | $d(\hat{\rho}_1(.))$ | $d(\hat{\sigma}(.))$ |
| MS                         | 0.01 (0.034)      | 0.027 (0.344)        | -0.012 (0.058)       |
| SET                        | -0.108*** (0.034) | -0.163 (0.343)       | -0.069 (0.058)       |
| ST                         | -0.168*** (0.034) | 0.911*** (0.349)     | -0.006 (0.059)       |
| KM                         | -0.183*** (0.033) | -0.155 (0.34)        | -0.128** (0.057)     |
| Observations               | 100840            |                      |                      |
| $R^2$ , Adjusted $R^2$     | 0.613, 0.613      |                      |                      |
| Resid. SE, F stat. p-value | 1.317, <0.001     |                      |                      |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

It is important to check if the estimated metrics generate similar results, as they are observable by the econometrician. This is done in Table 7.7. Unfortunately, the results vary widely. As noted in Figure 6.4, the models do not perfectly match the metrics, which blurs the relationship found in the previous table.

Table 7.7: Models and estimated RC metrics

| Model                      | Metric            |                      |                      |
|----------------------------|-------------------|----------------------|----------------------|
|                            | $d(\hat{\mu}(.))$ | $d(\hat{\rho}_1(.))$ | $d(\hat{\sigma}(.))$ |
| MS                         | -0.005*** (0.001) | 0.213*** (0.02)      | 0.11 (0.186)         |
| SET                        | -0.405 (0.275)    | -0.104*** (0.018)    | 0.148 (0.186)        |
| ST                         | 0.363*** (0.023)  | 0.181 (0.186)        | -0.41*** (0.032)     |
| KM                         | 0.041** (0.019)   | 0.14 (0.186)         | -0.495*** (0.032)    |
| Observations               | 88126             |                      |                      |
| $R^2$ , Adjusted $R^2$     | 0.296, 0.296      |                      |                      |
| Resid. SE, F stat. p-value | 1.017, <0.001     |                      |                      |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

Also recall, the metric estimation error varied across the unobservable RN, which means that these general relationships might not be too informative. Table 7.8 shows the values of interactions between models, metrics, and which parameter changes.<sup>9</sup> The omitted group is the interaction with  $\mu$ . For

<sup>9</sup>Controls include model, metric FEs, and interactions: metric<sup>2</sup> (with itself) and model · metric<sup>2</sup>.

example, a big ACF difference is only a bad indicator when the parameter changing was  $\sigma$ , not  $\rho_1$  (for SET and KM); average separation is often bad when  $\mu$  is fixed.

Table 7.8: Models and estimated RC metrics - RN interactions

| Metric                     | SGP      | Model             |                   |                  |                  |
|----------------------------|----------|-------------------|-------------------|------------------|------------------|
|                            |          | MS                | SET               | ST               | KM               |
| $d(\hat{\mu}(.))$          | $\rho_1$ | 0.025* (0.013)    | -0.039*** (0.012) | 0.107*** (0.023) | 0.011 (0.009)    |
|                            | $\sigma$ | 0.053*** (0.011)  | 0.025** (0.012)   | 0.113*** (0.023) | 0.05*** (0.009)  |
| $d(\hat{\rho}_1(.))$       | $\rho_1$ | -0.005 (0.008)    | -0.01 (0.006)     | -0.014* (0.007)  | 0.004 (0.015)    |
|                            | $\sigma$ | 0.004 (0.008)     | 0.05*** (0.006)   | 0.013 (0.008)    | 0.119*** (0.016) |
| $d(\hat{\sigma}(.))$       | $\rho_1$ | -0.036*** (0.006) | 0.053*** (0.011)  | 0.007 (0.015)    | -0.006 (0.014)   |
|                            | $\sigma$ | -0.017*** (0.006) | 0.061*** (0.01)   | 0.012 (0.014)    | 0.015 (0.013)    |
| Observations               |          | 88126             |                   |                  |                  |
| $R^2$ , Adjusted $R^2$     |          | 0.737, 0.737      |                   |                  |                  |
| Resid. SE, F stat. p-value |          | 1.012, <0.001     |                   |                  |                  |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

The results of this section could be used as a practical recommendation to the econometrician, e.g., “if your KM-based estimated regimes have high volatility separation, the model selection is sound, as the others often struggle with such series”. However, this is currently very limited because: (i) the metrics’ estimation error hides the true relationships between model and metrics; (ii) as noted in Section 6.1, the three metrics used are not enough to fully characterize each DGP, thus the results vary widely across the unobservable RN; (iii) while the use of RC metrics abstracts away from specific DGPs, the results are still conditional on the limited population used in this work.

Some strategies could be employed to improve the estimated metrics’ meaning, such as using weights proportional to how close each data point is to a regime change, considering other dispersion measures, or even silhouette-like scores instead of simple absolute differences. On the front of characterizing the DGPs, more metrics should be added, such as 3rd and 4th moments, more ACF lags, and more general distribution distance measures. Perhaps regime-unconditional and residual metrics – merging the usual model diagnostics in this analysis – could also improve the characterization. Overall, the present exercise, while not perfect, is a step towards a more general and informative way to characterize the DGPs and hopefully generate practical results for the econometrician.

### 7.3 Identification and performance

In the Models’ Fixed Effect section, we discussed the qualities of the models. It can be interesting to analyze separately how matching each of the characteristics of the DGPs relates to the RMSE of an estimation. This is done in Table 7.9. Recall the definition of  $\Delta$  provided at the beginning of this section: a positive coefficient means that a higher difference between the true and estimated values is associated with a higher RMSE.

There are counterintuitive relationships between  $R^2$ ,  $BME(r)$ ,<sup>10</sup> and RMSE: better fit, more error. These will be better analyzed below. The second column shows that  $\rho_1$  can have an adverse effect, as

<sup>10</sup>Binarized mean error, the proportion of observations where the estimated regime is different from the true one.



it did in fact have the least straightforward relationship with RGP, RN, and model interactions. A similar result is seen in the matching of metrics.

Table 7.9: RMSE and identification

|                             | RMSE              |                   |                   |
|-----------------------------|-------------------|-------------------|-------------------|
|                             | (1)               | (2)               | (3)               |
| $R^2$                       | 0.411*** (0.003)  | 0.441*** (0.003)  | 0.435*** (0.003)  |
| $BME(r)$                    | -0.098*** (0.003) | -0.089*** (0.003) | -0.092*** (0.003) |
| $\Delta\text{Switches}$     | 0.002 (0.003)     | 0.014*** (0.003)  | -0.002 (0.003)    |
| $\Delta\text{Duration}$     | -0.018*** (0.001) | -0.023*** (0.001) | -0.029*** (0.001) |
| $\Delta\mu$                 |                   | 0.029*** (0.003)  |                   |
| $\Delta\rho_1$              |                   | -0.003** (0.002)  |                   |
| $\Delta\sigma$              |                   | 0.109*** (0.001)  |                   |
| $\Delta d(\hat{\mu}(.))$    |                   |                   | 0.023*** (0.003)  |
| $\Delta d(\hat{\rho}_1(.))$ |                   |                   | -0.060*** (0.002) |
| $\Delta d(\hat{\sigma}(.))$ |                   |                   | 0.141*** (0.002)  |
| Constant                    | 1.507*** (0.003)  | 1.406*** (0.003)  | 1.473*** (0.004)  |
| Observations                | 92,001            | 91,960            | 82,922            |
| $R^2$                       | 0.231             | 0.282             | 0.286             |
| Adjusted $R^2$              | 0.231             | 0.282             | 0.286             |
| Residual Std. Error         | 0.780             | 0.753             | 0.768             |
| F Statistic p-value         | <0.001            | <0.001            | <0.001            |

Note:

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01

To properly analyze the first column, Table 7.10 stratifies the fit and regime effects by model. We observe that regime errors only have a negative relationship with no-RS and KM, two models that have some ‘disregard’ for the regime structure, while for the traditional models, smaller errors improve performance. This aligns with Dacco; Satchell (1999), who found that minor errors in the regime variable can lead to large forecasting errors. Apparently, KM is an exception to that rule, given the lack of parametric assumptions on the RGP. For the fit, the positive relationship still stands, meaning that fitting the regime is not a good guide for performance. The RF has the highest  $R^2$  coefficient, another indication of its overfitting. MS has one similar to no-RS, attesting to its flexibility, but not as small as KM.

To further analyze the effect of matching the metrics, Table 7.11 shows model-metric interactions. MS has a massive improvement when matching the average and decent results on volatility. SET is similar but overall matching matters less. For ST, matching the average matters less, but it is more important for ACF and volatility. KM is overall the most balanced, with a decent relationship with all metrics.

Again, it might be important to check the interaction with RN. This is done in Table 7.12. The same controls are used. The effects are generally different when  $\mu$  changes, but not so much across  $\rho_1$  or  $\sigma$ . Additionally, the interactions are overall smaller, indicating a smaller dependence on these non-observable characteristics of the DGPs.

These results could also be taken as practical recommendations, e.g., “if the econometrician has a belief about the true volatility dispersion, and the ST estimated doesn’t match it, it might be a problem”. A substitute for belief would be a more general measure of variation of the metric across

Table 7.10: RMSE and  $R^2$  across models

|                                   | RMSE             |                   |
|-----------------------------------|------------------|-------------------|
|                                   | (1)              | (2)               |
| $R^2 \cdot \text{No RS}$          | 0.749*** (0.009) |                   |
| $R^2 \cdot \text{MS}$             | 0.746*** (0.009) |                   |
| $R^2 \cdot \text{SET}$            | 0.851*** (0.010) |                   |
| $R^2 \cdot \text{ST}$             | 0.849*** (0.013) |                   |
| $R^2 \cdot \text{KM}$             | 0.467*** (0.009) |                   |
| $R^2 \cdot \text{RF}$             | 1.482*** (0.011) |                   |
| $BME(r) \cdot \text{No RS}$       |                  | -0.914*** (0.012) |
| $BME(r) \cdot \text{MS}$          |                  | 0.497*** (0.021)  |
| $BME(r) \cdot \text{SET}$         |                  | 0.329*** (0.018)  |
| $BME(r) \cdot \text{ST}$          |                  | 0.347*** (0.012)  |
| $BME(r) \cdot \text{KM}$          |                  | -0.310*** (0.016) |
| Observations                      | 120,850          | 100,840           |
| $R^2$                             | 0.288            | 0.072             |
| Adjusted $R^2$                    | 0.288            | 0.072             |
| Residual Std. Error               | 1.786            | 2.039             |
| F Statistic p-value               | <0.001           | <0.001            |
| Note: *p<0.1; **p<0.05; ***p<0.01 |                  |                   |

Table 7.11: RMSE and metrics identification across models

|                                          | Metric             |                       |                       |
|------------------------------------------|--------------------|-----------------------|-----------------------|
| Model                                    | $d(\hat{\mu}(.) )$ | $d(\hat{\rho}_1(.) )$ | $d(\hat{\sigma}(.) )$ |
| SET                                      | 0.737*** (0.02)    | 0.445*** (0.006)      | 0.359*** (0.008)      |
| ST                                       | 0.36*** (0.007)    | 0.14*** (0.003)       | 0.421*** (0.007)      |
| MS                                       | 0.228*** (0.005)   | 0.091*** (0.008)      | 0.428*** (0.008)      |
| Observations                             | 89065              |                       |                       |
| $R^2$ , Adjusted $R^2$                   | 0.441, 0.441       |                       |                       |
| Resid. SE, F stat. p-value               | 1.529, <0.001      |                       |                       |
| <i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 |                    |                       |                       |

Table 7.12: Models and estimated RC metrics - RN interactions

| Metric                     | SGP      | Model             |                   |                   |                  |
|----------------------------|----------|-------------------|-------------------|-------------------|------------------|
|                            |          | MS                | SET               | ST                | KM               |
| $d(\hat{\mu}(.))$          | $\rho_1$ | -0.067*** (0.013) | 0.012 (0.011)     | 0.127*** (0.021)  | -0.004 (0.009)   |
|                            | $\sigma$ | -0.008 (0.011)    | 0.099*** (0.011)  | 0.154*** (0.021)  | 0.046*** (0.008) |
| $d(\hat{\rho}_1(.))$       | $\rho_1$ | -0.006 (0.014)    | -0.025** (0.011)  | -0.182*** (0.014) | -0.039 (0.026)   |
|                            | $\sigma$ | 0.002 (0.013)     | 0.076*** (0.011)  | -0.093*** (0.015) | 0.073** (0.034)  |
| $d(\hat{\sigma}(.))$       | $\rho_1$ | -0.048*** (0.01)  | -0.04*** (0.015)  | -0.074*** (0.025) | 0.007 (0.027)    |
|                            | $\sigma$ | -0.028*** (0.009) | -0.074*** (0.015) | -0.143*** (0.025) | 0.028 (0.024)    |
| Observations               |          | 89065             |                   |                   |                  |
| $R^2$ , Adjusted $R^2$     |          | 0.736, 0.736      |                   |                   |                  |
| Resid. SE, F stat. p-value |          | 1.051, <0.001     |                   |                   |                  |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01

time, such as the SD of the rolling metric.<sup>11</sup> This interpretation is less useful since such beliefs are not common, and we have the same external validity issues as before.

There are other matchable characteristics of the models, such as the  $\tau$  for threshold models, the  $\gamma$  for ST, and transition probabilities for MS. They are not included here, as they are not directly comparable across models and do not generate interesting analysis.

## 7.4 Incorrect number of regimes

This section analyzes how the models perform when the assumed number of regimes  $\hat{S}$  is different from the true one  $S$ . In this analysis, underestimating the number of regimes means estimating a simple  $AR(1)$  with one of the two-regime RGPs, while overestimating means estimating some of the two-regime models with a series generator from a simple  $AR(1)$ . The results should be interpreted as such.

In Table 7.13, the overall effect of underestimation even shows an improvement.<sup>[^km-nors]</sup> With overestimation, we observe a positive result of 0.7.<sup>12</sup> Column (2) shows that all of the metrics' interactions are negative. This means that when we overestimate, if the regimes were highly different in terms of the metrics, we have an attenuation of the adverse effect of the misspecification. That is, it is very useful to be able to match these big differences, even if we need to use more regimes than might be necessary.

When we consider the opposite scenario in (3), the interactions are still negative. But again, this is carried by KM, as if we remove it from the sample, we get positive results throughout all the metrics. Column (4) shows that with underestimation, but similar regimes, we have an attenuation of the misspecification effect.

To stratify the effects by model, consider Table 7.14. As noted before, the ST model performs poorly with single-regime data, while KM actually benefits from it. MS is a little more robust than

<sup>11</sup>The prediction power of this measure, for the true metric dispersion, must be studied.

<sup>12</sup>Controls include metrics FEs, and interactions of the  $S$  comparison with models, plus metrics with models, RGP family, and RN parameter.

Table 7.13: Mis-specification of the number of regimes

|                                    | RMSE              |                   |                   |                   |
|------------------------------------|-------------------|-------------------|-------------------|-------------------|
|                                    | (1)               | (2)               | (3)               | (4)               |
| $\hat{S} < S$                      | −0.809*** (0.017) |                   | −0.350*** (0.021) | −0.494*** (0.023) |
| $\hat{S} > S$                      | 0.714*** (0.020)  | 1.981*** (0.028)  |                   |                   |
| $\hat{S} > S : d(\hat{\mu}(.))$    |                   | −1.927*** (0.029) |                   |                   |
| $\hat{S} > S : d(\hat{\rho}_1(.))$ |                   | −2.166*** (0.096) |                   |                   |
| $\hat{S} > S : d(\hat{\sigma}(.))$ |                   | −2.727*** (0.050) |                   |                   |
| $\hat{S} < S : d(\hat{\mu}(.))$    |                   |                   | −0.520*** (0.028) | 0.301*** (0.032)  |
| $\hat{S} < S : d(\hat{\rho}_1(.))$ |                   |                   | 0.002 (0.097)     | 0.426*** (0.111)  |
| $\hat{S} < S : d(\hat{\sigma}(.))$ |                   |                   | −0.343*** (0.047) | 0.517*** (0.054)  |
| Constant                           | 1.479*** (0.013)  | 1.183*** (0.013)  | 1.446*** (0.014)  | 1.580*** (0.017)  |
| Strat.                             | none              | none              | none              | no KM             |
| Observations                       | 81,970            | 81,970            | 81,970            | 61,128            |
| R <sup>2</sup>                     | 0.498             | 0.519             | 0.492             | 0.537             |
| Adjusted R <sup>2</sup>            | 0.498             | 0.519             | 0.492             | 0.537             |
| Residual Std. Error                | 0.938             | 0.918             | 0.943             | 1.005             |
| F Statistic p-value                | <0.001            | <0.001            | <0.001            | <0.001            |

Note:

\*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01

SET. Given the specific definition of overestimation, this result is not perfectly comparable with Janczura; Weron (2010), who found a perceived big sensitivity of  $\hat{S}$  in the MS model.

Table 7.14: Mis-specification of the number of regimes

|                            | RMSE              |
|----------------------------|-------------------|
| $\hat{S} > S : \text{MS}$  | 1.152*** (0.022)  |
| $\hat{S} > S : \text{SET}$ | 1.177*** (0.024)  |
| $\hat{S} > S : \text{ST}$  | 3.072*** (0.031)  |
| $\hat{S} < S : \text{KM}$  | −0.964*** (0.022) |
| Constant                   | 1.297*** (0.011)  |
| Observations               | 100,840           |
| R <sup>2</sup>             | 0.564             |
| Adjusted R <sup>2</sup>    | 0.564             |
| Residual Std. Error        | 0.888             |
| F Statistic p-value        | <0.001            |

Note:

\*p&lt;0.1; \*\*p&lt;0.05; \*\*\*p&lt;0.01

This analysis can be expanded to a more general notion of over- and under-specification, by generating series with several  $S > 1$  and estimating them with models with several  $\hat{S} > 1$ .

## 8 Conclusion

This work studied regime-switching models and DGPs from a learning perspective, demonstrating how the regimes' distributions can be used to analyze them and generate stylized facts relevant to forecast performance. The general and expandable framework and implementation, is a contribution in itself. I represent any RS DGP as a combination of a regime-generating process and a series-generating process, and I formalize regime-conditional metrics as functions that characterize the regime distributions. This makes it possible to discuss model behavior both in terms of labels (RGP and regime nature) and in terms of observable regime characteristics computed from  $(y, \hat{r})$ .

The framework was implemented with a constrained subset of its full capacity. I restrict attention to stationary Gaussian  $AR(1)$  SGPs, two regimes, MS, SET, and ST RGPs, and a minimal set of RC metrics, summarized via distance across regimes.

Some facts about the DGPs' regimes' distributions were presented. For the MS RGP, the metrics define a clear-cut profile: for  $\mu$  changes, only the average separates the regimes; for  $\rho_1$  changes, both the ACF and SD separate the regimes; and for  $\sigma$  changes, only the SD separates the regimes. For SET and ST, the interaction between the RGP and the RN creates more complex profiles, where all metrics are different across regimes, and more metrics would be required to fully identify the DGP. The asymmetric RGPs have a similar separation but require a larger sample size to establish it. With approximately 60 observations, the separation of these stationary  $AR(1)$  series with  $\sigma = 1$  converged. The estimated metrics do not perfectly match the true ones, and the errors' distribution varies depending on which parameter is changing. The RMSE distributions of observations with correctly identified regimes have slimmer tails, but not always, especially for the MS model.

Regarding the performance of the models, the MS model has a worse overall fixed effect (1.78), but its flexibility allows it to commit fewer egregious errors overall, performing better in asymmetric regimes and showing the smallest relationship between  $R^2$  and RMSE. The SET (1.69) and ST (1.63) models have better fixed effects and perform better for intercept changes. ST is more robust to small parameter changes. All of them perform worse than the KM (1.30), which has surprisingly good efficiency. The RF (1.59) is second to it, but partially because of its 'generic' hyperparameterization, exemplifying the risk of overfitting in complex models. The KM changes the least across stratifications, attesting to its strong approximator qualities; ST follows, given its relation to Neural Networks.

The baseline effect of mis-specifying the RGP is an RMSE increase of 0.52. The  $\sigma$  changes make forecasting harder, and this effect is compounded by mis-specification. In small RNs, mis-specifying the RGP is not too problematic. MS models suffer more, while ST suffers less. Misestimating an RS model in a non-RS DGP is disastrous, increasing RMSE by 3.53, especially for ST.

Mis-specification across RS RGPs is not as relevant, and no specific model-RGP pair has a different fixed effect than the correctly specified pair. Interactions of model-RGP-RN are similarly insignificant, but the RC metrics provide another way to characterize the DGP. SET, ST, and KM perform well with high average separation; ST seems to struggle with high ACF separation, and KM benefits from SD differentiation. MS does not show significant results. Unfortunately, these results change when we switch to estimated metrics, and vary widely across RNs, e.g. a big ACF difference is a bad scenario for ST when the  $\sigma$  parameter was the cause. Still, this analysis has potential and improvements are discussed in Section 8.1.

Matching each component of the DGPs has different relationships with the RMSE. Regime errors are positively related for the ST/SET/MS models, as shown in the literature, but not for the no-RS and KM models, given their disregard for a proper regime structure. The fit is not a good indicator of performance. MS shows a massive improvement when matching the average and decent results on volatility. SET is similar, but overall matching matters less. For ST, matching the average matters

less, but it is more important for ACF and volatility. KM is overall the most balanced, with a decent relationship with all metrics.

Finally, the effect of estimating one of the RS models in a no-RS DGP increases the RMSE by 0.7, but when the series are highly different across the RC metrics, this effect is attenuated, as the added power to explain the highly different regimes has its benefits. Estimating a simple  $AR(1)$  with an RS DGP is often beneficial, but when the regimes are highly different, it starts to be harmful.

## 8.1 Limitations and further research

**DGPs and models:** While the metrics somewhat abstract from the DGPs, the results are still conditional on the specific population of DGPs considered here. Expanding the set of RGPs would be the first improvement. This could be done while maintaining a balanced relationship between threshold-based and Markov-based processes, and discrete transitions versus smooth transition processes. For example, including MS-ST DGPs and the mixed models by Chang; Choi; Park (2017) and Wu; Chen (2007). At the same time, variations on the models would be interesting, such as Markov models with non-geometric distributions and thresholds with  $|y_{t-1}|$  and  $|\Delta y_{t-1}|$  as transition variables.

On another front, one could focus on the importance of the SGP, considering more complex models than the simple  $AR(1)$ , introducing non-stationarity. More complex RNs could include changes in the family of the errors' distribution or simultaneous changes in multiple parameters. Models that match the new RGPs should follow, but also other general-purpose models, such as Neural Networks, given their relationship to ST models. The properties of ensemble RS models could also be assessed.

**Metrics:** Different strategies for metric calculation can help with better identification between the estimated and the true values, e.g., adding stronger weights to observations close to the regime switches, and using different dispersion measures, such as silhouette-like measures or squared distances.

Additionally, it was observed that the set of three metrics considered was insufficient to fully characterize all the DGPs, which led to unobservable factors being relevant in the model-DGP analysis. A more comprehensive set of metrics could help address this, including, for example, third and fourth moments, more lags of the autocorrelation, and more general distribution distances.

**Additional analyses:** Finally, there were many other questions that the framework can handle but were left unexplored. All the information about the regimes' distributions was collapsed into the difference between them, measured by the dispersion of the RC metrics. However, the individual information of each regime can be useful, e.g., to analyze whether the models perform better in the regime with the high or low parameter value.

One could test whether common approaches to selecting the models' hyperparameters, such as choosing the number of regimes via AIC or BIC, work overall and in which scenarios they do not. More agnostic approaches to estimate what metrics change across regimes can help with practical recommendations. The dispersion of the rolling series of the metrics is one option.

A final interesting option is to include complex but no-RS models, such as models with frequent 'outliers' or non-linearities, and understand how well RS models capture these dynamics, similar to the work of Chib (1998). The knowledge presented in this work could be applied to real data, as a form of validation, and to test-drive the results.<sup>13</sup>

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<sup>13</sup>AI disclaimer: large language models were used sparingly as a research tool, for code autocompletions, and the final text review.

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## A DGPs, models, and metrics

### A.1 RGPs and models

#### A.1.1 No RS (no-RS)

**Hypothesis:** No regime switching, always at regime ‘1’.

$$\begin{aligned} r_t^1(.) &= 1 \\ r_t^s(.) &= 0, \quad s \in \{2, \dots, S\} \end{aligned} \tag{A.1}$$

**Empirical model:** a simple  $AR(1)$  model, estimated via OLS, with no regime-switching component.

#### A.1.2 Structural break (SB)

**Hypothesis:** Regime changes occur at specific time points  $\tau \in (1 : T)^{S-1}$ . Formally:

$$\begin{aligned} r_t^s(. ; \tau) &= \mathbb{1}(\tau'_{s-1} < t \leq \tau'_s), \quad \forall s \in \{1, \dots, S\} \\ \tau &\in \mathbb{N}^{S-1}, \quad \tau_s > \tau_{s-1} \quad \forall s, \quad \tau' = (0, \tau, T) \end{aligned} \tag{A.2}$$

**Empirical model:** Given  $\tau$ , the model estimates  $\mu$  and  $\rho_1$  via OLS in each regime.  $\tau$  is chosen by minimizing the sum of squared residuals over a grid search of breakpoints.

This approach is similarly defined by Bai; Perron (1998), with a review of other options provided by Casini; Perron (2018). Note that this RGP and model were excluded from the final analysis.

#### A.1.3 Self-exciting threshold (SET)

**Hypothesis:** Regime changes occur when the series, possibly at a lag  $d \in \mathbb{N}^*$ , crosses specific threshold values  $\tau \in \mathbb{R}^{S-1}$ . Transformations of the variable can also be considered.<sup>14</sup> Formally:

$$\begin{aligned} r_t^s(. ; (\tau, d, g)) &= \mathbb{1}(\tau'_{s-1} < g(y)_{t-d} \leq \tau'_s), \quad \forall s \in \{1, \dots, S\} \\ \tau &\in \mathbb{R}^{S-1}, \quad \tau_s > \tau_{s-1} \quad \forall s, \quad \tau' = (-\infty, \tau, \infty), \quad d \in \mathbb{N}^* \end{aligned} \tag{A.3}$$

**Empirical model:** Given  $\tau$  and  $d$ , the model estimates  $\mu$  and  $\rho_1$  via OLS in each regime.  $\tau$  and  $d$  are determined by minimizing the sum of squared residuals over a grid search of breakpoints and lags. Alternatively,  $d$  can be fixed.

This approach is similarly defined by Tong; Lim (1980), with a review of other options provided by Chen; Liu; So (2011).

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<sup>14</sup>For example,  $g(x) = |x|$  or  $g(x) = \Delta x$ .

#### A.1.4 Smooth transition (ST)

**Hypothesis:** Regime changes occur smoothly, governed by a continuous function  $g$ , often a CDF, based on the difference between the series and the threshold  $\tau \in \mathbb{R}$ , possibly at a lag  $d \in \mathbb{N}^*$ . Medeiros (2000) demonstrated that a generalization to  $S$  regimes is equivalent to a neural network with  $S$  nodes in the hidden layer, but this work considers only  $S = 2$ . Formally:

$$\begin{aligned} r_t^1(\cdot; (\tau, d, g)) &= g(y_{t-d} - \tau), & r_t^2(\cdot; (\tau, d, g)) &= 1 - r_t^1(\cdot; (\tau, d, g)) \\ \tau &\in \mathbb{R}, & d &\in \mathbb{N}^* \end{aligned} \quad (\text{A.4})$$

Often, the function  $g$  depends on a smoothness parameter  $\gamma$ , such that as  $\gamma \rightarrow \infty$ ,  $g \rightarrow \mathbb{1}$ . This parameter can be estimated jointly with the others.

**Empirical model:** Estimated via non-linear least squares of the residuals, optimizing over  $\mu$ ,  $\rho_1$  (for each regime),  $\tau$ , and  $\gamma$ . This process involves numerical optimization, which depends on starting values and does not guarantee a global optimum.

This approach is similarly defined by Teräsvirta (1994), with a review of other options provided by Dijk; Teräsvirta; Franses (2002).

#### A.1.5 Markov-Switching (MS)

**Hypothesis:** Regime changes occur stochastically, following a Markov process with transition matrix  $\Gamma \in [0, 1]^{S \times S}$ . The probability of being in regime  $s$  at time  $t$  depends only on the regime at time  $t - 1$ , with  $\Gamma$  often implying some persistence. Formally:

$$\begin{aligned} r_t^s(\cdot; \Gamma) &\sim P(r_t^s = 1 | r_{t-1}) =: \Gamma_{s, r_{t-1}} \\ \Gamma &\in [0, 1]^{S \times S}, \quad \sum_{i=1}^S \Gamma_{s, i} = 1 \quad \forall s \end{aligned} \quad (\text{A.5})$$

**Empirical model:** Various algorithms can be used, including maximum likelihood estimation, expectation-maximization, and Markov chain Monte Carlo methods. The EM algorithm employs Kalman filtering to find smoothed probabilities of  $r$ , updates conditional probabilities based on the current parameter estimates, and iteratively maximizes the likelihood until convergence.

This approach is similarly defined by Hamilton (1989), with a review of other options provided by Song; Woźniak (2021).

#### A.1.6 Unsupervised clustering (UC)

**Hypothesis:** Regime switching occurs, but no specific hypothesis is made about the RGP.

**Model:** Unsupervised clustering techniques, such as K-Means, can estimate regimes based on  $y_t$ , its lags, and rolling moments. Given the regimes,  $\mu$  and  $\rho_1$  are estimated via OLS. This hybrid approach yields non-standard asymptotic properties. Other clustering techniques could also be employed, but this work focuses on the general K-Means clustering problem, defined as follows:

$$\begin{aligned} \hat{r}_t^s(\cdot; (\text{norm}, \text{centroid})) &= \mathbb{1}(y_t \in R_s) \\ R &= \arg \min_{R'} \sum_{s=1}^{\hat{S}} \sum_{y_t \in R'_s} \text{norm}(y_t - \text{centroid}(R'_s)) \end{aligned} \quad (\text{A.6})$$

This approach is similarly defined by Akioyamen; Tang; Hussien (2020), with a review of additional clustering techniques provided by Paparrizos; Yang; Li (2024).

### A.1.7 Random forests (RF)

**Hypothesis:** There is no regime switching; instead, non-linearity is captured by the tree and ensemble structure of the RF.

**Model:** A random forest is estimated based on  $y_t$ , its lags, and rolling moments.

A review of the time series RF literature is provided by Hu; Szymczak (2022).

## A.2 Metrics

The estimated conditional mean and standard deviation can be calculated as, respectively:

$$\hat{\mu}(y, r|s) := \sum_{t=1}^T r_t^s \cdot y_t \quad (\text{A.7})$$

$$\hat{\sigma}(y, r|s) := \sqrt{\frac{1}{1 - \sum_{t=1}^T (r_t^s)^2} \sum_{t=1}^T r_t^s \cdot (y_t - \hat{\mu}(y, r|s))^2} \quad (\text{A.8})$$

Note the bias correction factor in the denominator of the RC SD, which is necessary given the estimated mean.

As noted, in the case of binary  $r_t$ , only the observations of regime  $s$  have non-zero weights, and the formulas are respectively equivalent to:

$$\frac{1}{|R_s|} \sum_{y_t \in R_s} y_t, \quad \sqrt{\frac{1}{|R_s| - 1} \sum_{y_t \in R_s} (y_t - \hat{\mu}(y|s))^2}$$

For a regime-conditional moment of  $(y_t, y_{t-j})$ , we must define the notion of ‘being in the same regime’. Consider  $r_t^s \cdot r_{t-j}^s$ , which has a correct ‘truth table’ for binary regimes, but also has an interpretation for continuous ones: when closer to 1, the higher the weight of both  $y_t$  and  $y_{t-j}$  being in regime  $s$ . But, this ignores that fact that  $y_t$  and  $y_{t-j}$  can be in the same regime, but in different regime instances. To account for that, the correct weighting should consider the whole window of  $y_{t-j}, \dots, y_t$ :

$$\hat{\rho}_j(y, r|s) = \frac{\sum_{t=1+j}^T \left( \prod_{k=1}^j r_k^s \right) \cdot (y_t - \hat{\mu}(y, r|s)) \cdot (y_{t-j} - \hat{\mu}(y, r|s))}{\sum_{t=1}^T \left( \prod_{k=1}^j r_k^s \right) \cdot (y_{t-j} - \hat{\mu}(y, r|s))^2} \quad (\text{A.9})$$

Note the absence of bias correction. While it could be present, it can generate larger-than-one correlations, and is often omitted.

For binary regimes, this is equivalent to calculating the unweighted autocorrelation of every concurrent window of regime  $s$ .

In this work, I am currently using the binary version of the RC metrics, calculated after binarizing  $r$ .

Recall that a RC metric returns a sequence with entries for each regime, so when describing, e.g., the RC mean  $\mu(y, r)$ , I am referring to:

$$\mu(y, r) := \mu(y, r|S) = (\mu(y, r|s))_{s \in 1:S} \quad (\text{A.10})$$

### A.2.1 True moments of the considered DGPs

Given the weakly stationary within regimes assumption, the regime-conditional moments are independent of the RGP, and are the simple  $AR(1)$  moments:

$$\begin{aligned} \mu(y_t|s) &\equiv E[y_t|y_t \in R_s] = \frac{\mu^s}{1-\rho_1^s} \\ \sigma(y_t|s) &\equiv \text{Var}[y_t|y_t \in R_s] = \sqrt{\frac{(\sigma^s)^2}{1-(\rho_1^s)^2}} \\ \rho_j(y_t|s) &\equiv \text{Corr}[y_t, y_{t-1}|y_t \in R_s] = (\rho_1^s)^j, \quad j \in \mathbb{N}^* \end{aligned} \quad (\text{A.11})$$

As described in Section 3.3.2, there can be better estimators for populational RC metrics than the ones defined in the above section. One can simply plug in the estimated parameters in the equation above to get a better estimator of the moments. In this work, I use this approach, which is also more computationally efficient.

## B Diagnostics

### B.1 Random errors

The Figure B.1 shows the correlation of the errors across the parallelization structure. A simple visual check shows no evident patterns and an overall low correlation, as expected.

The Figure B.2 shows the distribution of a size 3000 sample of the errors, via the usual histogram and QQ-plot. The distribution is very close to normal, as expected.

Figure B.1: Random errors - Correlation across parallelization structure

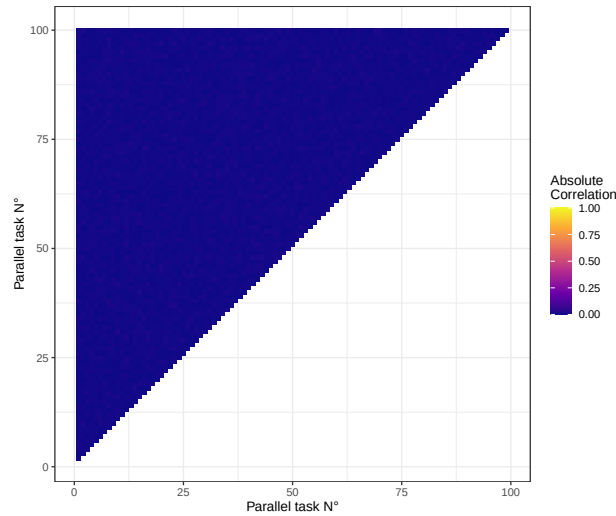
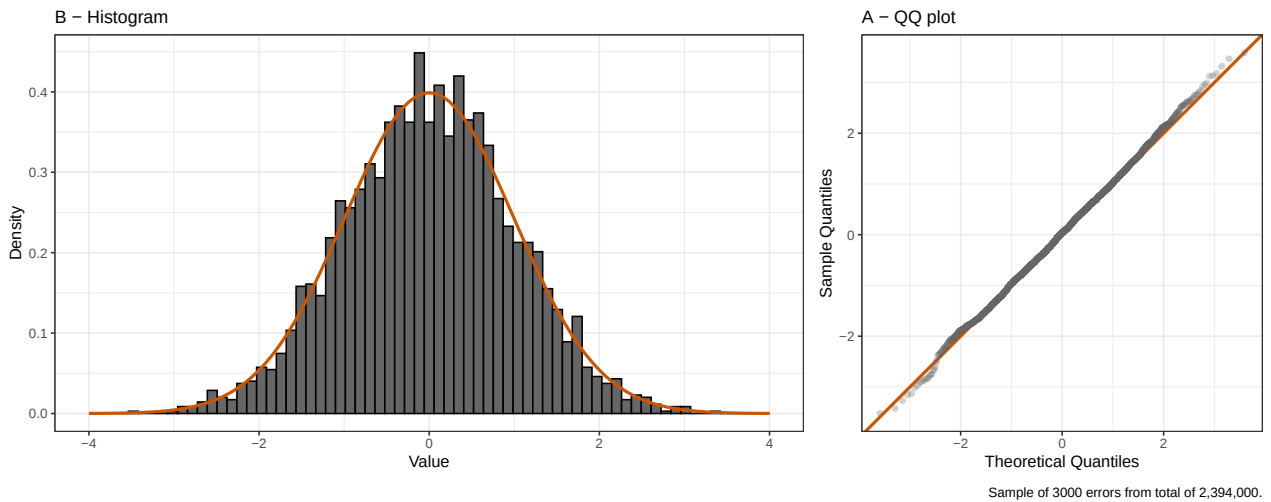


Figure B.2: Random errors - Distribution



### B.2 Estimated errors

The Figure B.3 presents the distribution of the estimation errors (residuals) across models, while Figure B.4 shows the distribution of forecasting errors. Overall, the distributions align with expectations, with the former exhibiting fatter tails than the latter. In the former, approximately 10,000 observations fall outside the range of the x-axis, some of which were identified as outliers.

Figure B.3: Residuals - Distribution

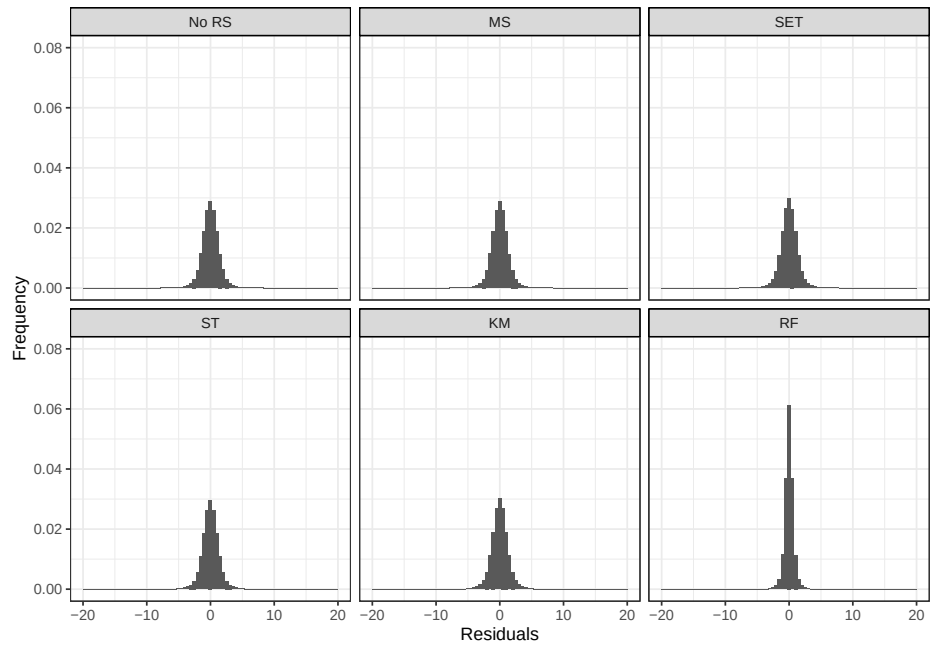
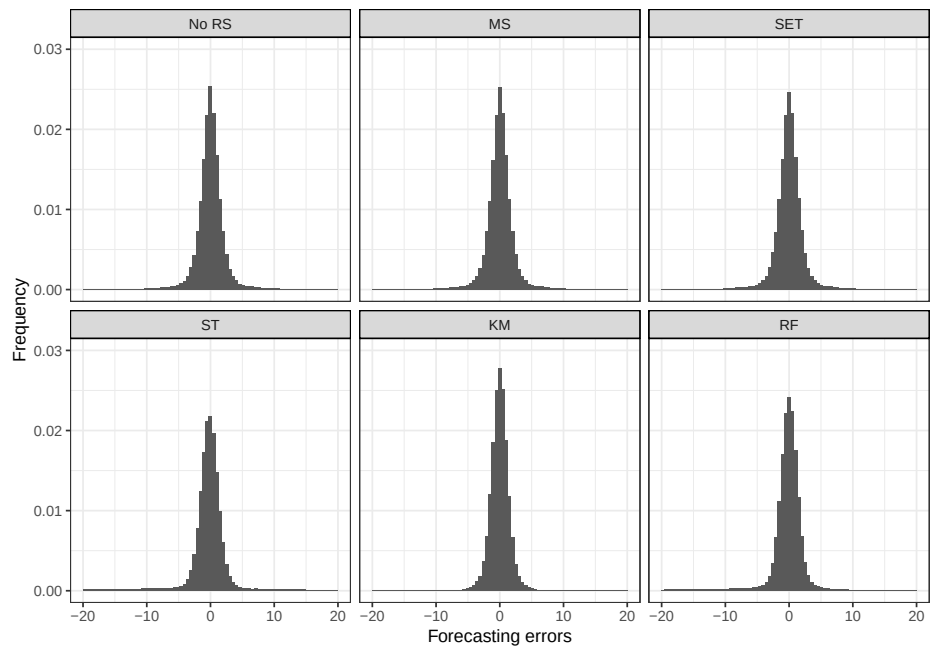


Figure B.4: Forecasting errors - Distribution

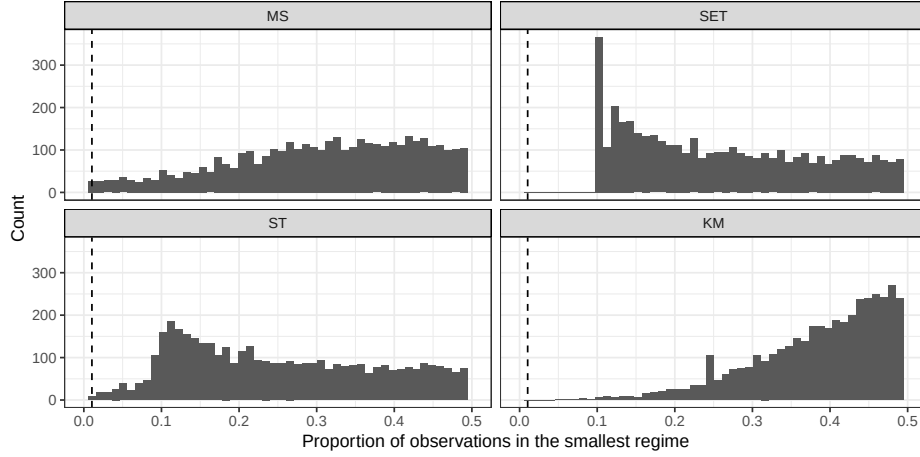




### B.3 Regime proportions

The Figure B.5 illustrates the distribution of the proportion of the least frequent estimated regime, separated by model. Estimations below the dashed line had only two observations in the regime and were excluded.

Figure B.5: Regime proportion - Distribution



### B.4 Parameters and model metadata

The Figure B.6 visualizes the distribution of the estimated parameters across models and parameters. All by-regime values of a parameter are grouped together in the same panel. Overall, the distributions align with expectations, with approximately 10,000 values falling outside the range of the x-axis, some of which were identified as outliers.

The Figure B.7 displays the distribution of RGP-related metadata, such as the MS transition probabilities, ST  $\gamma$ , and SET  $\tau$ . Overall, the distributions align with expectations.

The Table B.2 compares the model parameters to the true values of the DGPs. Each group of rows corresponds to the moments of a DGP. The first two columns relate to the values conditional on regimes 1 and 2, while the third column provides the unconditional values. Each cell contains the moment value, with the p-value of the null hypothesis that the moment equals its true value in brackets. The table includes only symmetric RGPs and large SGPs. Note that the moments do not need to match exactly, as all models allow for parameter variation, which differs from the assumptions about regime natures.

### B.5 Metrics calculation

The calculated metrics are presented in Table B.3. Each group of rows corresponds to the moments of a DGP. The first two columns relate to the values conditional on regimes 1 and 2, while the third column provides the unconditional values. Each cell contains the moment value, with the p-value of the null hypothesis that the moment equals its true value in brackets. The table includes only symmetric RGPs and large SGPs. Note that the moments do not need to match exactly, as all models allow for parameter variation, which differs from the assumptions about regime natures.

Figure B.6: Parameters - Distribution

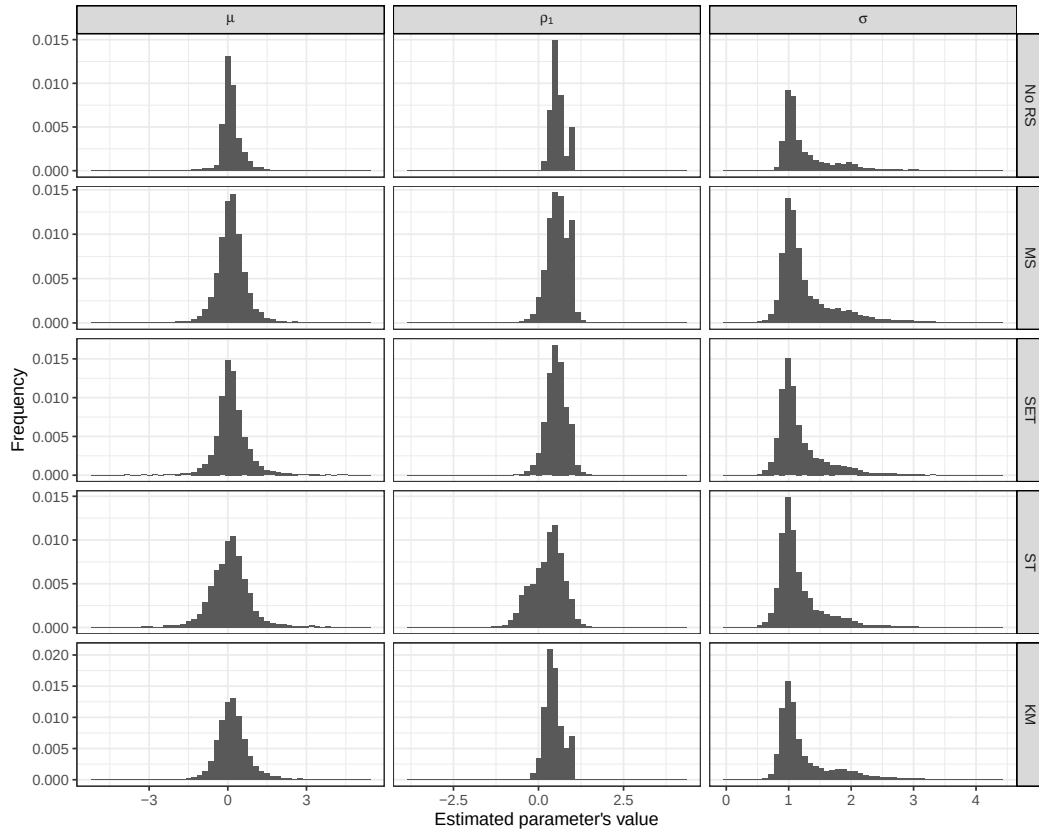


Figure B.7: Model metadata - Distribution

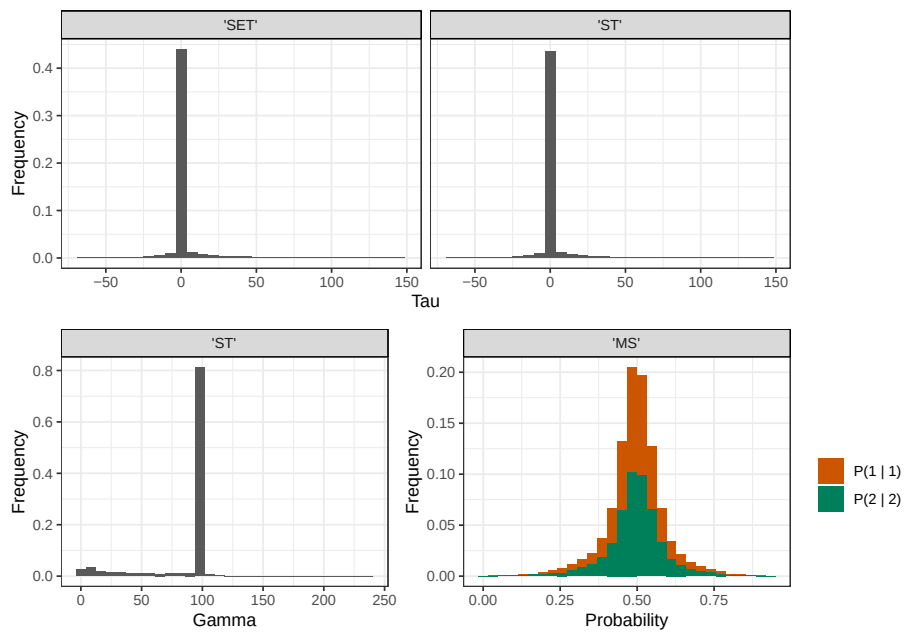


Table B.1: RMSE and simulation index relationship

|                                          | RMSE             |
|------------------------------------------|------------------|
| $i$                                      | -0.346 (4.293)   |
| $i^2$                                    | 1.578 (2.036)    |
| $i^3$                                    | 0.553 (1.444)    |
| $\log(i)$                                | 0.004 (0.014)    |
| Constant                                 | 1.284*** (0.073) |
| Observations                             | 123,137          |
| $R^2$                                    | 0.00002          |
| Adjusted $R^2$                           | -0.00001         |
| Residual Std. Error                      | 1.026            |
| F Statistic p-value                      | 0.622            |
| <i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 |                  |

Table B.2: Estimated coefficients across DGPs

| RGP                                      | RN                | $\mu^s$        |                | $\rho_1^s$     |               | $\sigma^s$     |                |
|------------------------------------------|-------------------|----------------|----------------|----------------|---------------|----------------|----------------|
|                                          |                   | $s = 1$        | $s = 2$        | $s = 1$        | $s = 2$       | $s = 1$        | $s = 2$        |
| MS, $p_{21} = .1$                        | $\mu (0, 1)$      | .16*** (.011)  | .6*** (.015)   | .68*** (.008)  | .57*** (.008) | 1.07*** (.007) | 1.07*** (.007) |
|                                          | $\rho_1 (.4, .6)$ | -.01 (.012)    | .03** (.011)   | .32*** (.011)  | .74*** (.008) | 1.04*** (.006) | 1.04*** (.006) |
|                                          | $\sigma (1, 2)$   | .01 (.016)     | -.01 (.017)    | .46** (.014)   | .49 (.014)    | 1.42*** (.011) | 1.7*** (.012)  |
| SET, $\tau = .5$                         | $\mu (0, 1)$      | .22*** (.018)  | 1.11*** (.028) | .74*** (.01)   | .5 (.014)     | .98*** (.006)  | .97*** (.006)  |
|                                          | $\rho_1 (.4, .6)$ | .29*** (.018)  | .04* (.021)    | .36*** (.01)   | .77*** (.009) | .97*** (.006)  | .96*** (.005)  |
|                                          | $\sigma (1, 2)$   | -.05* (.023)   | .02 (.024)     | .47** (.014)   | .5 (.013)     | 1.19*** (.008) | 1.46*** (.009) |
| ST, $\tau = .5$                          | $\mu (0, 1)$      | -.19*** (.027) | .77*** (.017)  | .17*** (.021)  | .11*** (.02)  | .97*** (.006)  | .95*** (.006)  |
|                                          | $\rho_1 (.4, .6)$ | -.13*** (.032) | .04 (.022)     | -.35*** (.018) | .71*** (.01)  | .97*** (.005)  | .96*** (.006)  |
|                                          | $\sigma (1, 2)$   | -.05 (.043)    | -.02 (.044)    | .26*** (.021)  | .25*** (.021) | 1.39*** (.009) | 1.64*** (.009) |
| <i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 |                   |                |                |                |               |                |                |

Table B.3: Estimated metrics across DGPs

| RGP               | RN                | $\hat{\mu}(. s)$ |                |             | $\hat{\rho}_1(. s)$ |               |            | $\hat{\sigma}(. s)$ |                |             |
|-------------------|-------------------|------------------|----------------|-------------|---------------------|---------------|------------|---------------------|----------------|-------------|
|                   |                   | $s = 1$          | $s = 2$        | $\perp s$   | $s = 1$             | $s = 2$       | $\perp s$  | $s = 1$             | $s = 2$        | $\perp s$   |
| MS, $p_{21} = .1$ | $\mu (0, 1)$      | .21*** (.012)    | 1.79*** (.011) | 1.03 (.014) | .38*** (.005)       | .38*** (.005) | .55 (.003) | 1.18*** (.007)      | 1.18*** (.006) | 1.41 (.006) |
|                   | $\rho_1 (.4, .6)$ | 0 (.008)         | .05** (.022)   | .03 (.012)  | .16*** (.006)       | .57*** (.004) | .51 (.004) | 1.02 (.005)         | 1.49*** (.012) | 1.31 (.008) |
|                   | $\sigma (1, 2)$   | -.01 (.012)      | 0 (.021)       | 0 (.013)    | .38*** (.005)       | .38*** (.005) | .44 (.003) | 1.2*** (.007)       | 2.22*** (.012) | 1.81 (.01)  |
| SET, $\tau = .5$  | $\mu (0, 1)$      | -.19*** (.014)   | 2.07*** (.009) | 1.47 (.016) | .2*** (.008)        | .37*** (.004) | .6 (.002)  | 1.05*** (.007)      | 1.11*** (.005) | 1.47 (.008) |
|                   | $\rho_1 (.4, .6)$ | -.08*** (.007)   | 1.29*** (.014) | .63 (.013)  | .08*** (.006)       | .39*** (.006) | .54 (.003) | 1*** (.005)         | 1.2*** (.007)  | 1.3 (.007)  |
|                   | $\sigma (1, 2)$   | -.43*** (.007)   | .85*** (.02)   | -.01 (.012) | .19*** (.005)       | .17*** (.007) | .44 (.003) | 1.1*** (.004)       | 2.06*** (.012) | 1.59 (.009) |
| ST, $\tau = .5$   | $\mu (0, 1)$      | 1.02*** (.006)   | .56*** (.008)  | .85 (.006)  | .12*** (.005)       | .04*** (.007) | .27 (.004) | 1.04*** (.004)      | 1.01*** (.005) | 1.05 (.003) |
|                   | $\rho_1 (.4, .6)$ | .41*** (.009)    | -.66*** (.011) | -.39 (.011) | -.01*** (.008)      | .45*** (.005) | .53 (.003) | 1*** (.006)         | 1.23*** (.007) | 1.27 (.007) |
|                   | $\sigma (1, 2)$   | .81*** (.01)     | -.59*** (.014) | 0 (.013)    | .1*** (.006)        | .24*** (.005) | .44 (.003) | 1.33*** (.006)      | 1.89*** (.009) | 1.81 (.007) |

Note: \*p<0.1; \*\*p<0.05; \*\*\*p<0.01